

Track A: The Bar Construction and the Genus Tower

1. The bar construction on algebraic curves

The classical bar construction of a graded algebra lives on the cofree tensor coalgebra $T^c(s^{-1}\bar{A})$; its differential encodes the multiplication. For chiral algebras on a curve X three features distinguish the theory. The multiplication is mediated by a logarithmic propagator $d\log(z_i - z_j)$, and the bar differential becomes an integral transform on the Fulton–MacPherson compactification of configuration spaces. The bar coalgebra carries two coproducts: a non-cocommutative deconcatenation on ordered configurations $\text{Conf}_n^<(\mathbb{R})$, producing the primitive E_1 -coalgebra $\bar{B}^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, and a cocommutative coshuffle on the Ran space, producing the symmetric factorization coalgebra $\bar{B}^\Sigma(\mathcal{A}) = \text{Sym}^c(s^{-1}\bar{\mathcal{A}})$ as its Σ_n -coinvariant shadow. On curves of genus $g \geq 1$ the propagator deforms via the prime form, the bar differential acquires curvature $\kappa(\mathcal{A}) \cdot \omega_g$ from the Hodge bundle, and the universal Maurer–Cartan element $\Theta_{\mathcal{A}}$ of the modular convolution algebra controls the genus tower.

The ordered bar is the primitive object of this monograph. The symmetric bar, and with it every statement about the modular characteristic κ and the shadow obstruction tower, is obtained as the image under a canonical averaging map $\text{av}: \bar{B}^{\text{ord}}(\mathcal{A}) \rightarrow \bar{B}^\Sigma(\mathcal{A})$ described in §1.6.

1.1. The 1-form

Let X be a smooth algebraic curve over $\mathbf{C}$. On the configuration space $C_2(X) = X \times X \setminus \Delta$, define the logarithmic 1-form

$$\eta_{12} = d \log(z_1 - z_2) = \frac{dz_1 - dz_2}{z_1 - z_2}. \quad (0.0.1)$$

This form has a simple pole along Δ with residue 1. The residue depends on no coordinate, no metric, no level. A double pole $dz/(z-w)^2$ transforms under coordinate change and has no coordinate-independent residue; a regular 1-form extracts nothing from the collision. The logarithmic derivative is the unique 1-form on $C_2(X)$ with a first-order pole along Δ and conformally invariant residue.

On $C_3(X)$, the three pullbacks $\eta_{12}, \eta_{23}, \eta_{31}$ satisfy the *Arnold relation*

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0, \quad (0.0.2)$$

the differential-form image of the partial-fractions identity $\frac{1}{(z_1-z_2)(z_2-z_3)} + \frac{1}{(z_2-z_3)(z_3-z_1)} + \frac{1}{(z_3-z_1)(z_1-z_2)} = 0$. Arnold proved that (0.0.2), replicated across all triples $\{i, j, k\} \subset \{1, \dots, n\}$, generates the complete ideal of relations in $H^*(\text{Conf}_n(\mathbf{C}), \mathbf{C})$.

Under $z \mapsto w(z)$ the propagator transforms as $\eta_{ij} \mapsto \eta_{ij} + d \log \frac{w(z_i) - w(z_j)}{z_i - z_j}$; near the diagonal this reduces to $d \log w'(z_i)$, the standard $\text{Diff}(X)$ cocycle. The propagator is a BRST ghost for diffeomorphisms.

1.2. Chiral algebras on a curve

Write $\mathcal{D}_X$ for the sheaf of differential operators on X . A *chiral algebra* $\mathcal{A}$ is a $\mathcal{D}_X$ -module equipped with a *chiral bracket*

$$\mu^{\text{ch}}: j_* j^*(\mathcal{A} \boxtimes \mathcal{A}) \longrightarrow \Delta_!(\mathcal{A}),$$

where $j: X \times X \setminus \Delta \hookrightarrow X \times X$ is the inclusion of the complement of the diagonal and $\Delta_! \mathcal{A}$ is the $!$ -pushforward along the diagonal; concretely $\Delta_! \mathcal{A} \cong \mathcal{A} \otimes \delta_\Delta$. The source $j_* j^*(\mathcal{A} \boxtimes \mathcal{A})$ consists of sections of the external product with poles of arbitrary order along Δ ; the chiral bracket extracts the singular part and projects it onto the diagonal.

In local coordinates the chiral bracket encodes the *operator product expansion*

$$a(z) b(w) \sim \sum_{n \geq 0} \frac{a_{(n)} b(w)}{(z - w)^{n+1}}.$$

The n -th products $a_{(n)}b \in \mathcal{A}$ are extracted by $a_{(n)}b = \text{Res}_{z=w}[a(z)b(w)(z-w)^n]$. The zeroth product is the Lie-type bracket; the first product carries the symmetric/metric data; higher products ($n \geq 2$) are determined by the $\mathcal{D}_X$ -module structure. The *Borcherds identity* couples all pole orders into a single constraint,

$$\sum_{j \geq 0} \binom{m}{j} (a_{(n+j)}b)_{(m+k-j)}c = \sum_{j \geq 0} (-1)^j \binom{n}{j} (a_{(m+n-j)}(b_{(k+j)}c) - (-1)^n b_{(n+k-j)}(a_{(m+j)}c)),$$

specialising at $n = 0$ to Jacobi, at $m = 0$ to the commutator formula, at $n = 1, m = k = 0$ to the derivation property of $a_{(0)}$.

1.3. The bar complex on X

Place sections $a_1, \dots, a_n \in \mathcal{A}$ at distinct points $z_1, \dots, z_n \in X$ and assign to each ordered pair (i, j) the form $\eta_{ij} = d \log(z_i - z_j)$. For a spanning tree T of the complete graph on $\{1, \dots, n\}$ with edge set $E(T)$ form

$$\omega_T(a_1, \dots, a_n) = a_1(z_1) \otimes \dots \otimes a_n(z_n) \otimes \bigwedge_{(i,j) \in E(T)} \eta_{ij},$$

a section of $\Omega_{\log}^{n-1}$ of degree $|E(T)| = n - 1$ on $\text{Conf}_n(X)$.

The Fulton–MacPherson compactification $\overline{\text{FM}}_n(X)$ resolves $\text{Conf}_n(X)$ by iterated real oriented blowup along all diagonal strata: a blowup, not a complement. Points of $\overline{\text{FM}}_n(X) \setminus \text{Conf}_n(X)$ record limiting direction and relative scale as points collide; boundary codimension-one faces are indexed by subsets $S \subseteq \{1, \dots, n\}$ with $|S| \geq 2$.

The *ordered chiral bar complex* on X (the primitive object):

$$\bar{B}_X^{\text{ord}}(\mathcal{A}) = \bigoplus_{n \geq 1} (s^{-1}\bar{\mathcal{A}})^{\otimes n} \otimes \Gamma(\overline{\text{FM}}_n(X), \Omega_{\log}^{n-1}).$$

Here $\bar{\mathcal{A}} = \ker(\varepsilon: \mathcal{A} \rightarrow \omega_X)$ is the augmentation ideal (using $\bar{\mathcal{A}}$ rather than $\mathcal{A}$ is essential:), and s^{-1} is desuspension with $|s^{-1}a| = |a| - 1$ in cohomological convention ($|d| = +1$). The deconcatenation coproduct

$$\Delta[s^{-1}a_1 | \dots | s^{-1}a_n] = \sum_{i=0}^n [s^{-1}a_1 | \dots | s^{-1}a_i] \otimes [s^{-1}a_{i+1} | \dots | s^{-1}a_n]$$

splits an ordered sequence at every position. The differential decomposes as

$$d_{\bar{B}} = d_{\text{int}} + d_{\text{res}} + d_{\text{dR}}.$$

The three components: d_{int} is the internal differential of $\mathcal{A}$; d_{res} extracts OPE residues along codimension-one boundary strata of $\overline{\text{FM}}_n(X)$ at the pole order dictated by $\bigwedge \eta_{ij}$; d_{dR} is the de Rham differential on the logarithmic forms.

1.4. $d^2 = 0$ and Arnold+Borcherds

Expand $(d_{\text{int}} + d_{\text{res}} + d_{\text{dR}})^2$. The diagonal terms: $d_{\text{int}}^2 = 0$, $d_{\text{dR}}^2 = 0$; the cross-terms $[d_{\text{int}}, d_{\text{dR}}] = 0$ by Leibniz. The remaining five terms pair into three cancellation mechanisms:

Disjoint supports: when $\{i, j\} \cap \{k, l\} = \emptyset$ the boundary faces are transverse, and the residue operators commute.

Shared index: when $\{i, j\} \cap \{k, l\} = \{j = k\}$ the triple collision $z_i = z_j = z_l$ lies in a codimension-two stratum. The form contribution $\eta_{ij} \wedge \eta_{jl}$ paired with its cyclic partners vanishes by the Arnold relation; the corresponding algebraic contractions $(a_{i(m)}a_j)_{(n)}a_l + (a_{j(m)}a_l)_{(n)}a_i + (a_{l(m)}a_i)_{(n)}a_j$ vanish by Borcherds. Arnold kills the form; Borcherds kills the algebra; both are consumed.

Same pair: $d_{\text{res}}^{(ij)} d_{\text{res}}^{(ij)} = 0$ by the desuspension sign rule.

Result: $d_{\bar{B}}^2 = 0$ iff the n -th products satisfy the Borcherds identity. The bar construction accepts exactly chiral algebras.

All of this is genus 0. At genus $g \geq 1$ the propagator acquires quasi-periodicity and $d_{\text{fb}}^2 \neq 0$: the curvature $d_{\text{fb}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$ produces a single scalar governing the genus expansion. Genus 0 is uncurved algebra; genus ≥ 1 is curved geometry.

1.5. The two dualities of the bar complex

The cobar functor $\Omega(\bar{B}(\mathcal{A}))$ is the categorical logarithm inverted on the *Koszul locus*: the full subcategory on which

$$\varepsilon: \Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$$

is a quasi-isomorphism. This is bar-cobar *inversion*, recovering $\mathcal{A}$ from its bar complex (Theorem B). The Koszulness question, which controls the scope of this inversion, is the central organizing property of the programme and is characterised by twelve equivalent conditions in §9.

A second, linear operation on bar cohomology produces the *Koszul dual algebra* $\mathcal{A}^!$. First, take bar cohomology $\mathcal{A}^{\text{!}} := H^*(B(\mathcal{A}))$, a graded coalgebra, and apply graded linear duality: $\mathcal{A}^! := (\mathcal{A}^{\text{!}})^{\vee}$. The cobar recovers the

original; linear duality of bar cohomology produces a different algebra except at self-dual fixed points (e.g. $\text{Vir}_{c=13}$).

A third, geometric operation acts on the bar coalgebra. Verdier duality on $\text{Ran}(X)$ sends the factorization coalgebra $\bar{B}_X^\Sigma(\mathcal{A})$ to the *homotopy Koszul dual* $\mathcal{A}_\infty^!$, a factorization *algebra* carrying full chain-level data. The cohomology of $\mathcal{A}_\infty^!$ recovers $\mathcal{A}^!$ on the Koszul locus; this identification is the content of Theorem A, the Verdier intertwining theorem.

On $X = \mathbb{A}^1$, classical Koszul duality (Priddy, Beilinson–Ginzburg–Soergel, Loday–Vallette) embeds: the affine line deformation-retracts to a point (but the retraction is additional data, not a canonical identification), the FM chain cooperad becomes quasi-isomorphic to the associative cooperad, and the bar construction collapses to the classical bar complex of the underlying Lie algebra. On $\mathbb{P}^1$, the global topology (Arnold relations, Fulton–MacPherson compactification) has no counterpart over a bare point.

1.6. E_1 primacy and the averaging map

The bar complex of a chiral algebra carries two operadic colours, each with its own Koszul duality.

The ordered bar (primitive). The Stasheff tensor bar $\bar{B}^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct is an E_1 -coalgebra: coassociative, not co-commutative. It is built on ordered configurations $\text{Conf}_n^<(\mathbb{R})$ parametrising positions of line operators on an interval; the linear ordering is retained, with no Σ_n -quotient. Its Koszul dual $\mathcal{A}_{\text{line}}^!$ governs the category of line operators: for affine Kac–Moody, $\mathcal{A}_{\text{line}}^!$ is the Yangian $Y(\mathfrak{g})$.

The symmetric bar (coinvariant shadow). The symmetric bar $\bar{B}^\Sigma(\mathcal{A}) = \text{Sym}^c(s^{-1}\bar{\mathcal{A}})$ is the Σ_n -coinvariant quotient of the ordered bar, with the coshuffle coproduct: 2^n bipartitions of an unordered set rather than $n+1$ consecutive splittings. This is an E_∞ -coalgebra, built on $\overline{\text{FM}}_n(\mathbb{C})$ with Σ_n -coinvariants: the Beilinson–Drinfeld factorization coalgebra on $\text{Ran}(X)$. Its Koszul dual $\mathcal{A}_{\text{ch}}^!$ is the Francis–Gaitsgory chiral Koszul dual via Verdier duality on $\text{Ran}(X)$.

The averaging map and R -matrix descent. The averaging map

$$\text{av}: \bar{B}^{\text{ord}}(\mathcal{A}) \longrightarrow \bar{B}^\Sigma(\mathcal{A})$$

takes Σ_n -coinvariants. For a chiral algebra with OPE poles, this descent is not naive: it uses the spectral R -matrix $R(z)$ as twisting datum. The

R -matrix is the degree-2 collision residue of the ordered Maurer–Cartan element,

$$r(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_{\mathcal{A}}^{E_1}),$$

a matrix-valued rational function encoding the full binary OPE data. Applying av at degree 2 collapses the profile to a single scalar shadow:

$$\begin{aligned} \text{av}(r(z)) &= \kappa(\mathcal{A}) && \text{for abelian and scalar families,} \\ \text{av}(r(z)) &= \kappa_{\text{dp}}(\mathcal{A}), \quad \kappa(\mathcal{A}) = \kappa_{\text{dp}}(\mathcal{A}) + \dim(\mathfrak{g})/2 && \text{for non-abelian affine Kac–Moody.} \end{aligned} \tag{0.0.3}$$

This is the modular characteristic package of Theorem D. At degree 3 the KZ associator projects to the cubic shadow $\mathfrak{C}$; at degree 4 the Yangian higher coproduct projects to the quartic resonance class $\mathfrak{Q}$. Each degree- n component of the ordered Maurer–Cartan element projects under av to the degree- n shadow of the modular obstruction tower $\Theta_{\mathcal{A}}$.

The five theorems as averaged invariants. The main theorems A–D and H are the invariants of the ordered Maurer–Cartan element $\Theta_{\mathcal{A}}^{E_1}$ that survive averaging. The modular characteristic κ , the shadow obstruction tower, the complementarity pairing, and the chiral Hochschild cohomology are all Σ_n -coinvariant projections of the richer E_1 -data. The ordered story is the primitive; the modular story is its shadow. The R -matrix is derived from the local OPE for vertex algebras (E_∞ -chiral) and is independent input for nonlocal quantum vertex algebras (E_1 -chiral, in the sense of Etingof–Kazhdan).

1.7. The Heisenberg atom

The Heisenberg chiral algebra $\mathcal{H}_k$ is generated by a single field α of conformal weight 1 with OPE $\alpha(z)\alpha(w) \sim k/(z-w)^2$. Only $\alpha_{(1)}\alpha = k \cdot \mathbf{1}$ is nonzero; the Borchers identity is satisfied trivially.

The ordered bar complex at tensor degree n is $(s^{-1}\alpha)^{\otimes n} \otimes \Gamma(\overline{\text{FM}}_n(X), \Omega_{\log}^{n-1})$. At degree 2, $d_{\text{res}}(s^{-1}\alpha \otimes s^{-1}\alpha \otimes \eta_{12}) = k \cdot \mathbf{1}$. At degree $n \geq 3$, every residue at a codimension-one collision stratum extracts $\alpha_{(1)}\alpha = k \cdot \mathbf{1}$; the remaining $n-2$ factors are α , and $\mathbf{1}_{(p)}\alpha = 0$ for $p \geq 0$ kills further contractions. All structure is captured at tensor degree 2: the bar complex is quadratic, and $\mathcal{H}_k$ is chirally Koszul. The modular characteristic is $\kappa(\mathcal{H}_k) = k$.

The Koszul dual is the curved symmetric chiral algebra

$$\mathcal{H}_k^! = (\text{Sym}^{\text{ch}}(V^*), m_0 = -k\omega),$$

a degree-2 curvature satisfying the curved A_∞ relation $m_1^2(a) = [m_0, a]$. The curvature is the bar-dual of the OPE coefficient.

The E_1 face of Heisenberg. The ordered bar carries the R -matrix $r(z) = k/z$, a scalar with a simple pole: the second-order OPE pole $k/(z-w)^2$ drops by one order under the $d \log$ kernel. Applying (0.0.3), $\text{av}(r(z)) = k = \kappa(\mathcal{H}_k)$. For Heisenberg the R -matrix *is* the modular characteristic: averaging loses nothing, because $r(z)$ is already Σ_2 -invariant. For the affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$, the R -matrix is the level-full Casimir form

$$r(z) = k \Omega / z, \quad \Omega = \sum_a J^a \otimes J_a$$

. Averaging collapses the Casimir to its trace:

$$\text{av}(k \Omega / z) = \frac{k \dim \mathfrak{g}}{2h^\vee} = \kappa_{\text{dp}}(\widehat{\mathfrak{g}}_k), \quad \kappa(\widehat{\mathfrak{g}}_k) = \kappa_{\text{dp}}(\widehat{\mathfrak{g}}_k) + \frac{\dim \mathfrak{g}}{2}.$$

The Casimir tensor structure, invisible to κ , is the data that builds the Yangian.

1.8. The Kac–Moody OPE

The affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$ has OPE

$$J^a(z)J^b(w) \sim \frac{f^{ab}{}_c J^c(w)}{z-w} + \frac{k\kappa^{ab}}{(z-w)^2},$$

with $f^{ab}{}_c$ the structure constants and κ^{ab} the Killing form normalised so long roots have squared length 2. The first pole carries the Lie bracket $J^a_{(0)}J^b = f^{ab}{}_c J^c$; the second carries the invariant form $J^a_{(1)}J^b = k\kappa^{ab} \cdot \mathbf{1}$. The bar differential decomposes as

$$\mu^{\text{ch}} = \underbrace{f^{ab}{}_c J^c \cdot \eta_{12}}_{\text{Lie bracket}} + \underbrace{k\kappa^{ab} \cdot \mathbf{1} \cdot d\eta_{12}}_{\text{inner product}}.$$

The Lie bracket contributes the tree-level bar differential (Chevalley–Eilenberg combinatorics); the inner product contributes the curvature κ . The coexistence of both poles, absent in the pure second-order Heisenberg, makes $\widehat{\mathfrak{g}}_k$ the first nontrivial example.

Remark 0.0.1 (The zero-dimensional shadow). When $X = \text{Spec } \mathbb{k}$ is a point, configuration spaces collapse, logarithmic forms vanish, and the chiral bar complex reduces to the classical bar construction of an associative algebra A . A quadratic algebra $A = T(V)/(R)$ has quadratic dual $A^\dagger = T(V^*)/(R^\perp)$; when A is Koszul, bar cohomology concentrates in diagonal degrees and

$H^*(B(A)) \cong (A^\dagger)^\vee$. The operadic instances $\text{Com}^\dagger = \text{Lie}$ and $\text{Sym}^\dagger = \Lambda$ recover the Chevalley–Eilenberg complex as a restricted bar construction. When Koszulness fails, the cobar-bar resolution $\Omega(B(A)) \xrightarrow{\sim} A$ is the universal replacement.

Everything above takes place at genus 0: the propagator is globally defined on $\mathbb{P}^1$, and the Arnold relation holds without correction. On a curve of genus $g \geq 1$, $z_1 - z_2$ has no global meaning; the curve is compact, and any meromorphic function with a simple zero must also have a pole. The Arnold relation acquires a defect, the bar differential no longer squares to zero, and the defect is controlled by a single scalar.

2. Curvature and the genus tower

2.1. The genus- g propagator

At genus $g \geq 1$, the replacement for $z_1 - z_2$ is the *prime form* $E(z_1, z_2)$: a section of $K^{-1/2} \boxtimes K^{-1/2}$ vanishing to first order along the diagonal with no other zeros. The bundle is $K^{-1/2} \boxtimes K^{-1/2}$; the prime form is a $(-\frac{1}{2}, -\frac{1}{2})$ -differential. In a local coordinate with diagonal $z_1 = z_2$,

$$E(z_1, z_2) = (z_1 - z_2)(1 + O(z_1 - z_2)) \cdot (dz_1)^{-1/2}(dz_2)^{-1/2}.$$

Analytic continuation around the B -cycle B_j in z_1 gives

$$E(z_1 + B_j, z_2) = -\exp(-\pi i \Omega_{jj} - 2\pi i \int_{z_2}^{z_1} \omega_j) \cdot E(z_1, z_2),$$

where $\Omega = (\Omega_{jk})$ is the period matrix and $\omega_1, \dots, \omega_g$ the holomorphic differentials normalised by $\oint_{A_j} \omega_k = \delta_{jk}$. The A -cycle monodromy is trivial.

The logarithmic derivative $\omega(z_1, z_2) := d_{z_1} \log E(z_1, z_2)$ is the fundamental meromorphic differential of the second kind. The *genus- g propagator*:

$$\eta_{12}^{(g)} = d \log E(z_1, z_2) + \pi \sum_{j,k=1}^g (\text{Im } \Omega)_{jk}^{-1} \text{Im} \left(\int_{z_2}^{z_1} \omega_j \right) \omega_k(z_1).$$

The period correction $(\text{Im } \Omega)_{jk}^{-1}$ compensates B -cycle monodromy, rendering $\eta_{12}^{(g)}$ single-valued; this is a full $g \times g$ matrix. The price: $\eta_{12}^{(g)}$ is no longer holomorphic. The propagator couples to the Hodge bundle $\mathbb{E} \rightarrow \overline{\mathcal{M}}_g$ through Ω .

2.2. Curvature

The Arnold relation at genus g acquires a defect. The period corrections produce a residual $(1, 1)$ -form, and the fibrewise bar differential satisfies

$$d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g, \quad (0.0.4)$$

where ω_g is the Arakelov $(1, 1)$ -form on the fibre of the universal curve $\pi: \mathcal{C}_g \rightarrow \overline{\mathcal{M}}_g$, and $\kappa(\mathcal{A}) \in \mathbb{k}$ is the *modular characteristic*, determined entirely by the leading OPE singularity.

For Heisenberg, every residue extracts $\alpha_{(1)}\alpha = k$ and $\kappa(\mathcal{H}_k) = k$. For Kac–Moody, the curvature receives the trace over the adjoint representation:

$$\kappa(\widehat{\mathfrak{g}}_k) = \frac{(k + h^\vee) \dim \mathfrak{g}}{2h^\vee}.$$

At the critical level $k = -h^\vee$ the shifted level vanishes and the bar complex is uncurved. The Sugawara construction is undefined (the central charge $c = k \dim \mathfrak{g} / (k + h^\vee)$ has a pole) but the curvature parameter vanishes.

For Virasoro at central charge c , the OPE is $T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}$. The bar propagator absorbs one pole order, so the collision residue is $r(z) = (c/2)/z^3 + 2T/z$ (cubic, not quartic:). The curvature computation gives

$$\kappa(\text{Vir}_c) = c/2.$$

The Koszul dual is Vir_{26-c} , with $\kappa(\text{Vir}_{26-c}) = (26 - c)/2$; the two sum to 13, not zero. At $c = 13$, Virasoro is self-dual under Koszul duality.

2.3. Period integrals restore nilpotence

The curvature (0.0.4) obstructs $d_{\text{fib}}^2 = 0$ but not the total $D_g^2 = 0$. Integration against A -cycle periods defines a connection on the family of bar complexes parametrised by $\overline{\mathcal{M}}_g$. The corrected total differential

$$D_g = d_{\text{fib}} + \nabla^{\text{GM}}$$

incorporates the Gauss–Manin connection on the Hodge bundle, and $D_g^2 = 0$: the $(1, 1)$ -curvature of the fibrewise differential is absorbed by the $(0, 1)$ -part of the base connection. The fibrewise class $\kappa \cdot \omega_g \in H^{1,1}(\mathcal{C}_g)$ is identified under the period map $\overline{\mathcal{M}}_g \rightarrow \mathcal{A}_g$ with the $(0, 1)$ -curvature of ∇^{GM} on $R^1\pi_*\mathbb{C}$, and the two curvatures cancel.

2.4. The genus tower (uniform weight)

The bar complex traces a family of curved cochain complexes over $\overline{\mathcal{M}}_g$. For *uniform-weight* algebras (single-generator, or multi-generator with all generators of the same conformal weight), the obstruction to flatness is the characteristic class

$$\text{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g \in H^*(\overline{\mathcal{M}}_g), \quad \lambda_g = c_g(\mathbb{E}), \quad (0.0.5)$$

where $\mathbb{E} = \pi_* \omega_\pi$ is the Hodge bundle: the vector bundle on $\overline{\mathcal{M}}_g$ whose fibre over $[C]$ is the g -dimensional space $H^0(C, \omega_C)$. At genus 1 the factorization holds unconditionally for all families. For *multi-weight* families at genus $g \geq 2$ the scalar formula *fails*: a cross-channel correction from mixed-propagator boundary graphs appears,

$$\text{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g + \delta F_g^{\text{cross}}, \quad (\text{all-weight}) \quad (0.0.6)$$

(Theorem ??). Every appearance of obs_g , F_g , or λ_g in this monograph carries explicit tagging: (UNIFORM-WEIGHT) or (ALL-WEIGHT, WITH CROSS-CHANNEL CORRECTION).

Grothendieck–Riemann–Roch applied to the universal curve $\pi: \mathcal{C}_g \rightarrow \overline{\mathcal{M}}_g$ computes the total obstruction: the Chern character of $R\pi_* \omega_\pi$ involves the Todd class of the relative tangent, the $\hat{A}$ -genus. The generating function in the uniform-weight lane:

$$\sum_{g \geq 1} \text{obs}_g(\mathcal{A}) \cdot x^{2g} = \kappa(\mathcal{A}) \cdot (\hat{A}(ix) - 1).$$

Since $\hat{A}(ix) = (x/2)/\sin(x/2) = 1 + x^2/24 + 7x^4/5760 + \dots$, the first Faber–Pandharipande coefficients are

$$F_1 = \frac{\kappa}{24}, \quad F_2 = \frac{7\kappa}{5760}, \quad F_3 = \frac{31\kappa}{967680},$$

with the general term $\kappa \cdot \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$. Each of these is the scalar-lane projection; for multi-weight families at $g \geq 2$, add $\delta F_g^{\text{cross}}$. Verify $F_1 = \kappa/24$ as sanity check, not $1/(2\pi)$.

The scalar $\kappa(\mathcal{A})$ is only the leading term. The bar complex carries higher-degree data: a universal Maurer–Cartan element $\Theta_{\mathcal{A}}$ whose projection to degree r yields a shadow invariant $S_r(\mathcal{A})$. On each primary deformation line L , the shadow generating function $H(t) = \sum_{r \geq 2} r S_r t^r$ satisfies the algebraic relation

$$H(t)^2 = t^4 Q_L(t), \quad Q_L(t) = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4) t^2, \quad (0.0.7)$$

a quadratic polynomial in three genus-0 invariants (Theorem ??). The infinite tower is algebraic of degree 2, governed by the discriminant $\Delta = 8\kappa S_4$.

The shadow obstruction tower is not merely formal. Its projections yield: the Pixton ideal of the modular curve, the integrable hierarchies of Gelfand–Dickey type, the Hitchin spectral curve of the shadow connection, the Baxter Q -operator, and the Bekenstein–Hawking entropy formula of three-dimensional gravity (Chapter ??). These are projections of one object, not separate results.

2.5. Complementarity

Verdier duality on the Ran space splits the center local system into two halves, one controlled by $\mathcal{A}$ and one by $\mathcal{A}^\dagger$, carrying a shifted-symplectic structure:

$$R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}}) \simeq \mathbf{Q}_g(\mathcal{A}) \oplus \mathbf{Q}_g(\mathcal{A}^\dagger).$$

What $\mathcal{A}$ sees as deformation, $\mathcal{A}^\dagger$ sees as obstruction. The natural pairing

$$\mathbf{Q}_g(\mathcal{A}) \otimes \mathbf{Q}_g(\mathcal{A}^\dagger) \rightarrow R\Gamma(\overline{\mathcal{M}}_g, \omega_{\overline{\mathcal{M}}_g}) \simeq \mathbb{k}[-(3g-3)]$$

is a $-(3g-3)$ -shifted symplectic form, and the two summands are Lagrangian: isotropic of half the total dimension. The scalar sum rule is family-dependent (never universal):

$$\kappa(\mathcal{A}) + \kappa(\mathcal{A}^\dagger) = \begin{cases} 0 & (\text{Kac–Moody, free field, lattice}), \\ 13 & (\text{Virasoro}), \\ 250/3 & (\mathcal{W}_3), \\ K_N \cdot (H_N - 1) & (\mathcal{W}_N). \end{cases}$$

The constant $\kappa + \kappa'$ is the Koszul conductor of the family; it is zero for KM and free fields, nonzero for $\mathcal{W}$ -algebras. Qualify the duality statement: “self-dual at $c = 13$ under Koszul duality,” never unqualified.

2.6. Feigin–Frenkel duality

The Sugawara field $T(z) = \frac{1}{2(k+h^\vee)} \sum_a :J^a(z)J^a(z):$ has central charge $c = k \dim \mathfrak{g} / (k + h^\vee)$. At the critical level $k = -h^\vee$ the denominator vanishes: $T(z)$ is undefined, not “ c diverges.” The chiral algebra acquires the Feigin–Frenkel center $\mathfrak{z}(\widehat{\mathfrak{g}}_{-h^\vee})$, isomorphic to functions on the space of ${}^L\mathfrak{g}$ -opers on the disc.

The Feigin–Frenkel involution $k \leftrightarrow -k - 2h^\vee$ is Koszul duality on configuration spaces: it interchanges $\mathcal{A}$ and $\mathcal{A}^!$ while preserving the bar-cobar adjunction. At the critical level, the Koszul dual is the Langlands-dual critical-level algebra $\widehat{L\mathfrak{g}}_{-Lh^\vee}$, not a renormalised copy of the original.

2.7. The total space

The universal configuration fibration $\pi_n: \overline{\mathcal{C}}_n(\mathcal{C}_g) \rightarrow \overline{\mathcal{M}}_g$ packages the bar complex into a single global object. Its boundary divisors carry two distinct geometric roles. *Collision divisors* (vertical): the fibrewise boundary of $\overline{\mathrm{FM}}_n(X_s)$, producing the bar differential d_{res} (OPE residues). *Degeneration divisors* (horizontal): the boundary of $\overline{\mathcal{M}}_g$ itself (nodal curves), producing the period corrections and Gauss–Manin terms.

Algebraic boundary (Borcherds) meets geometric boundary (period map and clutching). Their interaction is the curvature κ : the fibrewise bar differential fails to be nilpotent precisely because the vertical and horizontal boundaries do not commute.

3. Modular homotopy theory

The genus tower demands an organising framework. The fibrewise differential, the Gauss–Manin correction, and the clutching morphisms at boundary strata satisfy compatibility relations dictated by the stratification of $\overline{\mathcal{M}}_{g,n}$. The governing algebraic structure is the *modular operad*: an operad whose operations are indexed by connected stable graphs of arbitrary genus.

The *modular homotopy type* of $\mathcal{A}$ is the formal moduli problem classifying the inverse system of bar complexes,

$$\mathcal{M}_{\mathcal{A}}^{\mathrm{mod}}(R) := \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} \widehat{\otimes}_{\mathfrak{m}} R) / \sim$$

for local Artin rings R . Here $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ is the *modular convolution algebra*: a complete filtered cyclic L_∞ -algebra. Three inputs enter: a homotopy chiral algebra on the curve, a cooperadic chain complex on logarithmic configuration spaces, and a convolution L_∞ -structure mediating between them. The bar complex, the Koszul dual, the genus tower, the complementarity splitting, and the shadow obstruction tower are all representations of $\mathcal{M}_{\mathcal{A}}^{\mathrm{mod}}$.

The primitive story lives on ordered configurations. The E_1 convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}, E_1}$, built on ribbon graphs from $F\mathrm{Ass}$, is the primary object; the modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ is its image under the av map from ordered to symmetric. Every construction in this section begins with the E_1 structure and descends.

3.1. The modular operad

At genus 0, the combinatorics is governed by trees; at higher genus, by stable graphs with genus labels at their vertices. A *modular operad* (Getzler–Kapranov [?]) has operations indexed by connected stable graphs. A *connected stable graph* Γ has a label $g(v) \in \mathbb{Z}_{\geq 0}$ at each vertex satisfying $2g(v) - 2 + \text{val}(v) > 0$ and total genus $g(\Gamma) = b_1(\Gamma) + \sum_v g(v)$; trees are the genus-0 case. The structure maps are *separating gluing* $\xi_{\text{sep}}: \mathcal{O}(g_1, n_1+1) \otimes \mathcal{O}(g_2, n_2+1) \rightarrow \mathcal{O}(g_1+g_2, n_1+n_2)$ and *non-separating gluing* $\xi_{\text{ns}}: \mathcal{O}(g, n+2) \rightarrow \mathcal{O}(g+1, n)$.

A stable graph records a degeneration of a smooth curve into a nodal curve: vertices are irreducible components, edges are nodes, genus labels are component genera, half-edges are marked points. The set $\text{Gr}_{g,n}^{\text{st}}$ parametrises the boundary stratification of $\overline{\mathcal{M}}_{g,n}$. Every codimension-one boundary divisor is a one-edge degeneration; every codimension-two stratum is the intersection of exactly two codimension-one divisors, appearing with opposite orientations.

The *Feynman transform* replaces trees by stable graphs,

$$F\mathcal{O}(g, n) = \bigoplus_{\Gamma \in \text{Gr}_{g,n}^{\text{st}}} \frac{\hbar^{g(\Gamma)}}{|\text{Aut}(\Gamma)|} \bigotimes_{v \in V(\Gamma)} \mathcal{O}(g(v), \text{In}(v)). \quad (0.0.8)$$

3.2. The Feynman transform differential

The differential contracts one edge at a time,

$$d_{F\mathcal{O}}(\gamma_\Gamma) = \sum_{e \in E(\Gamma)} (-1)^{\epsilon(e)} \xi_e(\gamma_{\Gamma/e}),$$

with the sign from the position of e in the ordering on $E(\Gamma)$ recorded by $\det(E(\Gamma))$. The identity $d^2 = 0$ is the codimension-2 cancellation: each pair of edges contracts in either order, and the two terms cancel by associativity of $\mathcal{O}$. At genus 0, $F\mathcal{O}$ reduces to the operadic bar construction.

When $\mathcal{O} = \text{Com}$, each vertex factor in (0.0.8) is one-dimensional, and the Feynman transform becomes the stable graph complex

$$F\text{Com}(g, n) = \bigoplus_{\Gamma \in \text{Gr}_{g,n}^{\text{st}}} \frac{\hbar^{g(\Gamma)}}{|\text{Aut}(\Gamma)|} \det(E(\Gamma)).$$

When $\mathcal{O} = \text{Ass}$, the Feynman transform $F\text{Ass}$ sums over ribbon graphs (fatgraphs); this is the E_1 -primitive input for the ordered convolution algebra.

3.3. The convolution ladder

At each level of generality, the bar-type construction is a convolution dg Lie algebra whose Maurer–Cartan elements classify algebraic structures:

$$\begin{array}{lll}
\text{algebra :} & \text{Hom}(B(\text{Ass}), \text{End}_A) & A_\infty\text{-structures on } A \\
\text{operad :} & \text{Hom}_\Sigma(B(\mathcal{P}), \text{End}_V) & \mathcal{P}_\infty\text{-structures on } V \\
\text{modular :} & \text{Hom}_\Sigma(F\mathcal{O}, \text{End}_{\bar{B}(\mathcal{A})}) & \text{consistent genus towers}
\end{array} \tag{0.0.9}$$

The first line is topological Hochschild; the second is the Ginzburg–Kapranov deformation complex; the third governs this book. The Lie bracket is the convolution bracket

$$[f, g] = \mu \circ (f \otimes g) \circ \Delta - (-1)^{|f||g|} \mu \circ (g \otimes f) \circ \Delta,$$

with Δ the cooperadic decomposition and μ the operadic composition.

The *primitive E_1 convolution*

$$\mathfrak{g}_{\mathcal{A}}^{\text{mod}, E_1} := \text{Hom}_\Sigma(F\text{Ass}, \text{End}_{\bar{B}^{E_1}(\mathcal{A})})$$

is the ordered, ribbon-graph input: the algebraic form of the 't Hooft $1/N$ expansion. Setting $\hbar = 1/N^2$, the MC element $\Theta_{\mathcal{A}}^{E_1} \in \text{MC}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}, E_1})$ is the genus expansion $\sum_g N^{2-2g} F_g(\lambda)$ summed over ribbon graphs weighted by the genus of their embedding surface. Planar ($g = 0$) diagrams dominate at large N ; the shadow obstruction tower $\Theta^{E_1, \leq r}$ is the truncation to r -point 't Hooft interactions.

The *modular convolution algebra* is the av-image,

$$\mathfrak{g}_{\mathcal{A}}^{\text{mod}} := \text{Hom}_\Sigma(F\text{Com}, \text{End}_{\bar{B}(\mathcal{A})}). \tag{0.0.10}$$

The coinvariant map $F\text{Ass} \rightarrow F\text{Com}$ projects $\Theta^{E_1} \mapsto \Theta$: colour-ordered amplitudes to full amplitudes. On each primary line, the shadow tower is algebraic of degree 2: the entire infinite sequence is the Taylor expansion of $\sqrt{Q_L}$ for a single quadratic Q_L , the shadow metric (§??).

Remark (Homotopy types in three stages). The three levels of the convolution ladder mirror three levels of homotopy theory. *Over a field* (the bar complex of a graded algebra): $B(A) = (T^c(s^{-1}\bar{A}), d_B)$ determines the homotopy type of an associative algebra; formality corresponds to intrinsic formality; Massey products measure the obstruction. *Over a manifold*: the Sullivan minimal model $(\Lambda V, d)$ determines the rational homotopy type; Kähler manifolds are formal. The Heisenberg algebra is the vertex-algebraic

analogue: its shadow obstruction tower terminates at degree 2, the Gaussian archetype. *Over a curve at all genera:* the A_∞ -products m_k of the first stage become the modular L_∞ -brackets $\ell_k^{(g)}$; formality becomes the Gaussian archetype; the shadow obstruction tower is the modular Massey product tower. The shadow depth classification (§6) makes this analogy precise: the four classes G, L, C, M correspond to four levels of modular formality.

Koszulness as central question. Within this formality classification, the Koszulness property is the central organizing question of the programme. A chiral algebra is *chirally Koszul* iff its bar cohomology concentrates in diagonal degrees and the cobar-bar counit $\Omega(\bar{B}(\mathcal{A})) \rightarrow \mathcal{A}$ is a quasi-isomorphism. Twelve equivalent conditions (eight genuinely independent bidirectional equivalences among (i)–(x), with (vi) a proved consequence of (v) and (viii) a one-way chiral Hochschild consequence on the Koszul locus; one conditional on perfectness; one one-directional) characterise this locus. The Koszulness programme (§9) proves the core equivalences from PBW concentration, bar-cobar inversion, chiral Hochschild polynomial growth, and Koszul-complex acyclicity. All standard families are Koszul; shadow depth classifies the complexity of Koszul algebras, not Koszulness itself.

3.4. Homotopy chiral algebras

Fix an affine cover $\mathfrak{U} = \{U_\alpha\}$ of X . The Čech totalisation

$$A_\infty^{\text{ch}} := \check{C}^\bullet(\mathfrak{U}; \mathcal{A}) = \bigoplus_{p \geq 0} \prod_{\alpha_0 < \dots < \alpha_p} \mathcal{A}(U_{\alpha_0} \cap \dots \cap U_{\alpha_p})$$

carries a Ch_∞ -algebra structure (Malikov–Schechtman): the chiral bracket restricted to overlaps satisfies Jacobi up to a coherent system of higher homotopies, the *secondary Borchers operations*

$$F_n: \text{Lie}(n) \otimes \mathcal{Y}(n) \rightarrow P_n(\{A_\infty^{\text{ch}}\}, A_\infty^{\text{ch}}), \quad n \geq 2,$$

encoding derived OPE data at all pole orders. F_2 is the chiral bracket; F_3 is the Jacobiator homotopy arising from $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$, whose boundary has three points corresponding to OPE compositions $(a_{(m)}b)_{(n)}c$ and cyclic permutations. Higher operations F_n are determined inductively by $C_*(\overline{\mathcal{M}}_{0,n+1})$; the transferred operations are the associahedral A_∞ -corrections.

Different affine covers yield quasi-isomorphic Ch_∞ -algebras. The strict chiral algebra $\mathcal{A}$ is the H^0 of this structure: the primitive local object is the homotopy chiral algebra, and the strict algebra appears after rectification. Vallette rectification: the homotopy category of Ch_∞ -algebras is equivalent to the homotopy category of strict chiral algebras, via the bar-cobar adjunction.

3.5. Logarithmic Fulton–MacPherson spaces

The classical FM compactification records collisions on a smooth variety. Its logarithmic extension (Mok) records collisions along a divisor and provides the normal-crossings boundary structure on which $D^2 = 0$ depends.

For a simple-normal-crossing pair (X, D) , Mok constructs $\text{FM}_n(X|D)$: a smooth variety with corners compactifying $\text{Conf}_n(X \setminus D)$, with boundary stratification indexed by *rigid planted-forest types*. A planted-forest type ρ consists of a partition of $\{1, \dots, n\}$ into blocks (the roots), a rooted tree per block recording the hierarchy of collisions, and a grid-depth assignment $w_v \in \mathbb{Z}_{\geq 0}$ at each vertex recording the order of tangency with D . The ordinary FM is the case $D = \emptyset$.

The *degeneration formula*: for a semistable $W \rightarrow B$ with singular fibre $W_0 = \bigcup Y_v$ and a rigid type ρ ,

$$\text{FM}_n(W/B)(\rho) \longrightarrow \prod_{v \in V(S_\rho)} \text{FM}_{I_v}(Y_v|D_v)$$

is a proper birational correspondence factoring the global FM geometry into a fibre product of local pieces. Tropicalisation: $\text{Trop}(\text{FM}_n(\mathbb{C}|D)) = G_{\text{pf}}$, the planted-forest poset with grafting as partial order. The planted forest poset is the combinatorial skeleton of the geometric input, just as the stable graph complex is the skeleton of the Feynman transform.

The *Mok codimension formula*: for a type with grid depths $(w_1, \dots, w_r)$ and tree vertex counts $(|V(T_j)|)_{j=1, \dots, s}$,

$$\text{codim}(W_\rho) = \sum_{i=1}^r w_i + \sum_{j=1}^s (|V(T_j)| - 1).$$

For FM_3 : four codimension-one strata $(\{12\}, \{23\}, \{13\}, \{123\})$, three codimension-two strata $(\{12 < 123\}, \{23 < 123\}, \{13 < 123\})$.

The chain complexes $\mathcal{C}_{X|D}^{\log \text{FM}}(n) := C_\bullet(\text{FM}_n(X|D))$ carry cooperadic structure: the inclusion of boundary strata defines cocomposition maps. For a rigid stable graph Γ ,

$$\Delta_\Gamma^{\log} := (\nu_\Gamma)_* \circ \text{Res}_{D_\Gamma^{\log}} : \mathcal{C}_{\text{mod}}^{\log \text{FM}}(g, n) \rightarrow \bigotimes_{v \in V(\Gamma)} \mathcal{C}_{\text{mod}}^{\log \text{FM}}(g(v), I_v \sqcup H(v))[-|E_{\text{int}}(\Gamma)|],$$

where ν_Γ is the Mok birational correspondence and $\text{Res}_{D_\Gamma^{\log}}$ is the residue along the logarithmic boundary divisor.

3.6. The convolution L_∞ -algebra

A twisting morphism $\alpha: \mathcal{C} \rightarrow \mathcal{P}$ (satisfying $\partial\alpha + \alpha \star \alpha = 0$) determines a convolution sL_∞ -algebra $\text{hom}_\alpha(\mathcal{C}, \mathcal{P})$ (Robert-Nicoud–Wierstra, after Vallette) with underlying space $\prod_{n \geq 1} \text{Hom}_{\Sigma_n}(\mathcal{C}(n), \mathcal{P}(n))$. The brackets

$$\ell_k(f_1, \dots, f_k)(x) = \sum \gamma_{\mathcal{P}}((\alpha \circ 1)(f_1 \otimes \dots \otimes f_k) \Delta_{\mathcal{C}}^{(k)}(x))$$

come from iterated cooperadic decomposition. ℓ_1 is the convolution differential; ℓ_2 is the Lie bracket of coderivations; higher ℓ_k , $k \geq 3$, arise whenever $\mathcal{C}$ is not concentrated in a single degree.

The sL_∞ -algebra extends to ∞ -morphisms in either slot separately but not both simultaneously: the convolution is functorial in each slot, but does not assemble into a bifunctor. The MC3 categorical lift proceeds one slot at a time. The bar-cobar adjunction is a Quillen equivalence (Vallette); different Ch_∞ presentations yield quasi-isomorphic deformation theories.

3.7. Assembly

The modular convolution L_∞ -algebra:

$$\mathfrak{g}_{\mathcal{A}}^{\text{mod}} := \text{Conv}_\infty(\mathcal{C}_{\text{mod}}^{\log \text{FM}}, \text{End}_{\text{Ch}_\infty}(A_\infty^{\text{ch}})). \quad (0.0.11)$$

The cooperadic input is the log-FM chain cooperad $\mathcal{C}_{\text{mod}}^{\log \text{FM}}$; the operadic input is the Ch_∞ -endomorphism operad of A_∞^{ch} . The E_1 cousin $\mathfrak{g}_{\mathcal{A}}^{\text{mod}, E_1}$ uses the ribbon-graph-indexed cooperad and the ordered bar endomorphism operad.

One-slot convention: fix the log-FM (resp. ribbon) cooperad as the “geometric” slot; vary the chiral algebra input as the “algebraic” slot by ∞ -morphisms.

The *strict chart*: the coderivation dg Lie algebra

$$\mathfrak{g}_{\mathcal{A}}^{\text{mod}, \log}(\mathfrak{U}) := \text{Conv}_{\text{str}}(\mathcal{C}_{\text{mod}}^{\log \text{FM}}, \text{End}_{\text{Ch}_\infty}(A_\infty^{\text{ch}}))$$

has classical commutator bracket, no higher L_∞ brackets, but the same Maurer–Cartan moduli. The strict chart suffices for constructing $\Theta_{\mathcal{A}}$ and computing shadows; the full L_∞ structure is needed for transfer, formality, and gauge equivalence.

3.8. The modular bar functor

In log-FM coordinates,

$$B_{\text{mod}}^{\log}(\mathcal{A})(g, n) = \bigoplus_{\Gamma \in \text{Gr}_{g,n}^{\text{st}}} \left(\bigotimes_{v \in V(\Gamma)} \bar{B}^{\text{ch}}(A_{\infty}^{\text{ch}}; \text{In}(v)) \otimes C_{\bullet}(\text{FM}_{\Gamma}^{\log}) \otimes \overline{E_{\text{int}}(\Gamma)} \right)_{\text{Aut}(\Gamma)}.$$

The ingredients: at each vertex the transferred Ch_{∞} -bar contribution on its incoming half-edges; at each internal edge the propagator $P_{\mathcal{A}} = H_{\mathcal{A}}^{-1}$; the geometric factor $C_{\bullet}(\text{FM}_{\Gamma}^{\log})$; the orientation line $\overline{E_{\text{int}}(\Gamma)} = \det(\mathbb{k}E(\Gamma)) \otimes \det(H_1(\Gamma; \mathbb{k}))^{-1}$; the automorphism quotient.

3.9. The six-component differential

The total differential has six components, whose mutual cancellations constitute $D^2 = 0$:

$$D_{\text{mod}}^{\log} = \underbrace{d_{\check{C}}}_{\check{\text{Cech}}} + \underbrace{d_{\text{Ch}_{\infty}}}_{\text{transferred chiral}} + \underbrace{d_{\text{coll}}}_{\text{collision}} + \underbrace{d_{\text{sew}}}_{\text{separating}} + \underbrace{d_{\text{pf}}}_{\text{planted-forest}} + \underbrace{\hbar \Delta}_{\text{genus-raising}}.$$

$d_{\check{C}}$ is the Čech differential of the cover, acting on Čech indices. $d_{\text{Ch}_{\infty}}$ is the transferred chiral differential on each vertex, from secondary Borchers operations. d_{coll} contracts an internal edge by extracting the OPE residue at the corresponding codimension-one collision stratum of $\overline{\text{FM}}_n(X)$. The three positive-genus boundary operators act on a pure tensor $a \otimes \eta$ indexed by Γ :

$$\begin{aligned} d_{\text{sew}}(a \otimes \eta) &= \sum_{e \in E_{\text{sep}}(\Gamma)} \pm \mu_e(a) \otimes (\xi_e)_* \text{Res}_{D_e^{\log}}(\eta), \\ d_{\text{pf}}(a \otimes \eta) &= \sum_{F \in \text{PF}^{\text{rig}}(\Gamma)} \pm \mu_F(a) \otimes (\xi_F)_* \text{Res}_{D_F^{\log}}(\eta), \\ \hbar \Delta(a \otimes \eta) &= \hbar \sum_{e \in E_{\text{ns}}(\Gamma)} \pm \text{Tr}_e(a) \otimes (\xi_e)_* \text{Res}_{D_e^{\log}}(\eta). \end{aligned}$$

Separating edges give the clutching morphism (gluing two lower-genus curves); non-separating edges give the genus-raising self-gluing (BV operator); planted-forest strata give the genuinely logarithmic corrections from Mok. Explicitly,

$$d_{\text{pf}} = \sum_{\rho \in \text{PF}^{\text{rig}}} \epsilon_{\rho}(\kappa_{\rho})_* \text{pr}_{\rho}^* \otimes \mu_{\rho} :$$

Mok's degeneration formula internalised as a summand of the total differential.

3.10. $D^2 = 0$

The nilpotence $(D_{\text{mod}}^{\log})^2 = 0$ holds by three cancellation mechanisms. *Vertex labels:* $d_C^2 = 0$, $d_{\text{Ch}_\infty}^2 = 0$, and $\{d_C, d_{\text{Ch}_\infty}\} = 0$ by the Ch_∞ -algebra axioms on A_∞^{ch} . *One-edge expansions:* anticommutators of a vertex differential with an edge-contraction differential vanish by the Leibniz rule for coderivations. *Codimension-two cancellation:* the sum of all products of two distinct edge-contraction operators vanishes, because the codimension-two boundary of Mok’s log-FM compactification is a normal-crossings divisor (Mok, Theorem 3.3.1) and each codimension-two face appears as the intersection of exactly two codimension-one faces with opposite orientations.

The third mechanism is the nontrivial one. The normal-crossings property is essential: without it, codimension-two faces could have excess intersection, and the cancellation would fail.

4. The universal Maurer–Cartan element

Master conjectures MC1 through MC4 are *proved* (PBW concentration, MC element existence, thick generation in every type, completion tower). MC5 is partially proved: the analytic HS-sewing package is established at all genera, while the genuswise BV/BRST/bar identification remains conjectural; at genus 0 the algebraic BRST/bar comparison is proved (Theorem ??) and the tree-level amplitude pairing is conditional on Corollary ?. Theorem A (bar-cobar adjunction), Theorem B (bar-cobar inversion on the Koszul locus), Theorem C (complementarity: C1 unconditional, C2 (UNIFORM-WEIGHT)), Theorem D (obstruction $\text{obs}_g = \kappa\lambda_g$ (UNIFORM-WEIGHT); multi-weight correction $\delta F_g^{\text{cross}}$), and Theorem H (chiral chiral Hochschild concentration $\{0, 1, 2\}$, $\dim \leq 4$) are the five Σ_n -coinvariant projections of a single E_1 object.

4.1. Γ -amplitudes and Taylor coefficients

Each stable graph produces a multilinear operation on the convolution algebra, built from chiral operations at vertices, propagators along edges, and geometric factors from log-FM strata. For $\Gamma \in \text{Gr}^{\text{st}}$ with $|V(\Gamma)| = k$ vertices and elements $f_1, \dots, f_k \in \mathfrak{g}_A^{\text{mod}}$, the Γ -*amplitude* assembles

$$\ell_\Gamma(f_1, \dots, f_k) = \mu_\Gamma \circ (\alpha_\Gamma^{\text{mod}} \otimes f_1 \otimes \dots \otimes f_k) \circ \Delta_\Gamma^{\log},$$

a k -linear map of cohomological degree $2 - 2g(\Gamma) - k$. Here μ_Γ is the graph composition (transferred chiral operations at vertices, propagators along

edges), $\alpha_\Gamma^{\text{mod}}$ is the modular twisting morphism component, and $\Delta_\Gamma^{\text{log}}$ is the cooperadic cocomposition.

The Taylor coefficients of the modular L_∞ structure sum over stable graphs of fixed genus and vertex count:

$$\ell_k^{(g)} = \sum_{\substack{\Gamma \in \text{Gr}^{\text{st}}, |V(\Gamma)|=k \\ b_1(\Gamma) + \sum_v g(v)=g}} \frac{1}{|\text{Aut}(\Gamma)|} \ell_\Gamma.$$

The homotopy Jacobi identity is encoded by $D^2 = 0$ on the total convolution algebra.

Low-order Taylor coefficients. At tree level, the binary bracket is the transferred Lie bracket: $\ell_2^{(0)}(\alpha, \beta) = \pi[m_2(\iota(\alpha), \iota(\beta))]$. The ternary bracket sums over the three binary trees in Trees_3 (channels s, t, u), each inserting a homotopy h at the internal edge. The quaternary bracket sums over the fifteen labeled trees in Trees_4 (two unlabeled topologies: three balanced $((ab)(cd))$ and twelve caterpillar $((((ab)c)d))$), each with two propagator insertions; this is the classical homological perturbation lemma in its L_∞ incarnation.

The first genus-raised bracket is unary, the BV operator on the single graph with one vertex and one self-loop:

$$\ell_1^{(1)}(\alpha) = \text{tr}(P_{\mathcal{A}} \circ \alpha \circ \delta^{\text{ns}}) = \sum_i \langle e^i, \alpha(e_i) \rangle_{\text{Sh}},$$

trace over the propagator contracting two half-edges at the non-separating node of $\overline{\mathcal{M}}_{1,1}$. The genus-one binary bracket

$$\ell_2^{(1)}(\alpha, \beta) = \text{tr}(P_{\mathcal{A}} \circ m_2(\iota\alpha, -) \circ \delta^{\text{ns}} \circ \iota\beta) + (\alpha \leftrightarrow \beta)$$

accounts for the separating node of $\overline{\mathcal{M}}_{1,2}$. These low-order brackets are the only data needed for shadow computations through degree 4.

4.2. The bar-intrinsic construction

$\Theta_{\mathcal{A}}$ requires no equation to solve: it is extracted from a differential that already squares to zero.

Decompose the total bar differential $D_{\mathcal{A}}: \text{B}^{\text{mod}}(\mathcal{A}) \rightarrow \text{B}^{\text{mod}}(\mathcal{A})$ as $D_{\mathcal{A}} = d_0 + \Theta_{\mathcal{A}}$, where $d_0 = d_{\text{int}} + [\tau, -]$ is the tree-level local part. The *universal Maurer–Cartan element* is the remainder,

$$\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0 = \sum_{\Gamma \in \text{Gr}^{\text{st}}} \frac{W_\Gamma^{\text{log FM}}}{|\text{Aut}(\Gamma)|} \otimes \Phi_\Gamma^{\mathcal{A}} \in \text{MC}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}}), \quad (0.0.12)$$

with $W_{\Gamma}^{\log \text{FM}} \in H^*(\overline{\mathcal{M}}_{g(\Gamma),n(\Gamma)}^{\log})$ the log-FM fundamental class of the stratum and $\Phi_{\Gamma}^{\mathcal{A}} \in \text{Hom}(\overline{\mathcal{A}}^{\otimes n(\Gamma)}, \mathbb{k})$ the chiral amplitude (propagators on internal edges, operations at vertices, cyclic trace at the output).

The MC property is immediate. $D_{\mathcal{A}}^2 = 0$ by the normal-crossings property of $\overline{\mathcal{M}}_{g,n}^{\log}$; $d_0^2 = 0$ by construction; expanding $(d_0 + \Theta_{\mathcal{A}})^2 = 0$ gives

$$[d_0, \Theta_{\mathcal{A}}] + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0.$$

The MC property is a formal consequence of $\partial^2 = 0$ on the moduli space of curves, requiring no simple-Lie restriction, no convergence hypothesis, and no choice of basis. This is MC2 (conjecture-turned-theorem) in its cleanest form.

The E_1 analogue $\Theta_{\mathcal{A}}^{E_1}$ is constructed identically on the ordered, ribbon-graph-indexed convolution algebra, and $\Theta_{\mathcal{A}} = \text{av}(\Theta_{\mathcal{A}}^{E_1})$ under the coinvariant map $F\text{Ass} \rightarrow F\text{Com}$.

4.3. $\Theta_{\mathcal{A}}$ as universal modular twisting morphism

The Maurer–Cartan space of the modular convolution L_{∞} algebra is identified with the space of modular twisting morphisms,

$$\text{MC}_{\bullet}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}}) \simeq \text{Tw}_{\alpha^{\text{mod}}}(\mathcal{C}_{\text{mod}}^{\log \text{FM}}, \text{End}_{\text{Ch}_{\infty}}(A_{\infty}^{\text{ch}})).$$

Under this identification $\Theta_{\mathcal{A}}$ is the *universal modular twisting morphism*, the operadic analogue of the canonical flat connection on a principal bundle. A twisting morphism $\alpha: \mathcal{C} \rightarrow \mathcal{P}$ determines a twisted composition product $\mathcal{C} \circ_{\alpha} \mathcal{P}$ whose differential squares to zero iff α satisfies the MC equation. $\Theta_{\mathcal{A}}$ is the unique twisting morphism that simultaneously twists the modular bar complex and intertwines with the log-FM boundary structure.

The principal-bundle analogy is exact. The modular bar bundle $B_{\text{mod}}^{\log}(\mathcal{A}) \rightarrow \overline{\mathcal{M}}_{g,n}$ is the principal bundle; $\Theta_{\mathcal{A}}$ is the connection; the MC equation is flatness; the shadow tower (§6) is the Chern–Weil theory. The structure group is the pro-unipotent MC gauge group $\exp(\mathfrak{g}_{\mathcal{A}}^{\text{mod},0})$.

4.4. Tridegree, depth, and convergence

Each coefficient of $\Theta_{\mathcal{A}}$ carries a natural tridegree

$$(g, n, d) = (\text{loop genus}, \text{external degree}, \text{log boundary depth}),$$

with $d(\rho) = \#E_{\text{sep}}(\Gamma_\rho) + \#E_{\text{ns}}(\Gamma_\rho) + \text{depth}_{\text{pf}}(\Gamma_\rho)$ the codimension of the log-FM stratum. The depth filtration $F^d \mathfrak{g}_{\mathcal{A}}^{\text{mod}} = \bigoplus_{d' \geq d} (\mathfrak{g}_{\mathcal{A}}^{\text{mod}})_{*,*,d'}$ makes the convolution algebra complete:

$$\mathfrak{g}_{\mathcal{A}}^{\text{mod}} = \varprojlim_d \mathfrak{g}_{\mathcal{A}}^{\text{mod}} / F^d \mathfrak{g}_{\mathcal{A}}^{\text{mod}}.$$

The inverse limit $\Theta_{\mathcal{A}} = \varprojlim_r \Theta_{\mathcal{A}}^{\leq r}$ converges unconditionally: no analytic input, no growth estimate, no restriction on the algebra. Convergence is purely algebraic, guaranteed by completeness of the depth filtration. This is MC4 (completion tower): proved.

At fixed (g, n) , only finitely many stable graph types contribute. The depth filtration refines this further: at depth d , the stratum has codimension d in $\overline{\mathcal{M}}_{g,n}^{\text{log}}$, and the amplitude has internal-edge count bounded by d .

4.5. The weight filtration

Decompose the MC equation by internal-edge count,

$$\Theta_{\mathcal{A}} = \sum_{r \geq 1} \Theta_{\mathcal{A}}^{[r]}, \quad \Theta_{\mathcal{A}}^{[r]} = \sum_{\substack{\Gamma \in \text{Gr}^{\text{st}} \\ |E_{\text{int}}(\Gamma)|=r}} \frac{W_{\Gamma}^{\text{log FM}}}{|\text{Aut}(\Gamma)|} \otimes \Phi_{\Gamma}^{\mathcal{A}}.$$

Weight 1: $[D_{\text{loc}}, \Theta_{\mathcal{A}}^{[1]}] = 0$. Weight N : $[D_{\text{loc}}, \Theta_{\mathcal{A}}^{[N]}] + \frac{1}{2} \sum_{i+j=N} [\Theta_{\mathcal{A}}^{[i]}, \Theta_{\mathcal{A}}^{[j]}] = 0$. The obstruction to extending from weight $N-1$ to weight N lies in $H^2(\mathfrak{g}_{\mathcal{A}}^{\text{mod}}/F^{N+1}, D_{\text{loc}})$. The bar-intrinsic construction solves all stages simultaneously.

Weight 1: boundary divisors. In degree $(0, 4)$: the three boundary points of $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$ correspond to the three channels $(12|34)$, $(13|24)$, $(14|23)$,

$$\Theta_{\mathcal{A}}^{[1]}|_{(0,4)} = [\delta_s] \otimes \Phi_s + [\delta_t] \otimes \Phi_t + [\delta_u] \otimes \Phi_u.$$

The weight-one equation $\Phi_s + \Phi_t + \Phi_u = F_3$ identifies the Jacobiator homotopy with a boundary in the bar complex: the first Borchers identity. In degree $(1, 1)$: the single boundary divisor δ_{irr} gives $\Theta_{\mathcal{A}}^{[1]}|_{(1,1)} = [\delta_{\text{irr}}] \otimes \Delta$ with $\Delta = \sum_i e_i \otimes e^i$ the Casimir. In degree $(0, 5)$: the ten boundary divisors of $\overline{\mathcal{M}}_{0,5}$ (Petersen graph) recover the pentagon identity. The entire A_{∞} hierarchy is the tree-level projection of a single MC element.

Weight 2: the first planted-forest layer. At weight 2, codimension-two strata enter: classical products of two boundary divisors plus genuinely

logarithmic planted-forest strata with no DM counterpart. The weight-two equation is $[D_{\text{loc}}, \Theta_{\mathcal{A}}^{[2]}] = -\frac{1}{2}[\Theta^{[1]}, \Theta^{[1]}]$, and the planted-forest correction $\Theta_{\text{pf}}^{[2]}$ records how collision limits in $\text{FM}_n(X|D)$ correct the naive square. For the $\mathcal{W}_3$ central-parameter line, cubic gauge triviality kills the weight-2 obstruction; the first genuine invariant appears at weight 3.

4.6. The all-genus recursion

Projecting the MC equation to fixed (g, n) ,

$$d\Theta_{\mathcal{A}}^{(g,n)} + \frac{1}{2} \sum_{\substack{g_1+g_2=g \\ n_1+n_2=n+2}} \ell_2^{(0)}(\Theta_{\mathcal{A}}^{(g_1,n_1)}, \Theta_{\mathcal{A}}^{(g_2,n_2)}) + \Delta_{\text{ns}} \Theta_{\mathcal{A}}^{(g-1,n+2)} + \sum_{k \geq 3} \frac{1}{k!} \ell_k^{(\text{tree})}(\Theta_{\mathcal{A}}, \dots, \Theta_{\mathcal{A}})^{(g,n)} = 0.$$

Four terms: the internal differential, separating clutching, non-separating BV, and planted-forest corrections. This recursion determines $\Theta_{\mathcal{A}}^{(g,n)}$ inductively from lower (g', n') in lexicographic order. The base cases are $\Theta_{\mathcal{A}}^{(0,3)}$ (the chiral bracket) and $\Theta_{\mathcal{A}}^{(0,4)}$ (the Jacobiator data). The recursion terminates at each (g, n) .

The clutching identity. On each rigid boundary stratum ρ ,

$$\Theta_{\mathcal{A}}|_{\rho} = (\kappa_{\rho})_* \left(\bigotimes_{v \in V(S_{\rho})} \Theta_{\mathcal{A},v} \right).$$

The amplitude on the degenerate curve equals the product of amplitudes on the irreducible components, sewn by propagators along the nodes. This is the MC-level sewing axiom, and the starting point for the clutching law of the quartic shadow (§6).

Every invariant in this monograph is a projection of $\Theta_{\mathcal{A}}$: the curvature κ , the spectral discriminant $\Delta_{\mathcal{A}}$, the shadow metric Q_L , the sewing amplitudes, the $\hat{A}$ -genus, the Fredholm determinant, and the quartic contact invariant Q^{contact} . The five main theorems are five projections.

Projection of $\Theta_{\mathcal{A}}$	Invariant	Section
Scalar ($r = 2$, genus tower)	$\kappa(\mathcal{A})$, F_g , $\hat{A}$ -genus	§2
Binary genus-0 residue	classical r -matrix, Yangian seed	§7.7
Binary Verdier dual	complementarity $Q_g \oplus Q_g^!$	§2
Bar-cobar counit	inversion $\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$	§1
Chiral Hochschild	$\text{ChirHoch}^*(\mathcal{A})$, deformation ring	§9
Spectral discriminant	$\Delta_{\mathcal{A}}(x)$, genus-1 Hessian	§5
Cubic shadow ($r = 3$)	gauge-trivial obstruction	§6
Quartic resonance ($r = 4$)	$\mathfrak{Q}^{\text{contact}}$, clutching law	§6
MC tautological descent	shadow CohFT, $R^*(\overline{\mathcal{M}}_{g,n})$	§5
Primitive kernel	shell equations, branch BV	§5
Dirichlet–sewing lift	$S_{\mathcal{A}}(u)$, Euler–Koszul class	§8
Swiss-cheese colour decomposition	α_T , six projections (Vol. II)	§10

These five theorems are five projections of $\Theta_{\mathcal{A}}$. Theorem A constructs the arena in which $\Theta_{\mathcal{A}}$ lives. Theorem B recovers $\mathcal{A}$ from $\Theta_{\mathcal{A}}$ on the Koszul locus. Theorem C splits the genus tower into Lagrangian summands (C1 unconditional, C2 (UNIFORM-WEIGHT)). Theorem D extracts the scalar κ with $\hat{A}$ -genus generating function (uniform-weight; multi-weight adds $\delta F_g^{\text{cross}}$). Theorem H identifies the deformation ring via ChirHoch^* concentration.

The finite-order projections form an obstruction tower $\Theta_{\mathcal{A}}^{\leq 2} \rightarrow \Theta_{\mathcal{A}}^{\leq 3} \rightarrow \Theta_{\mathcal{A}}^{\leq 4} \rightarrow \cdots$ whose generating function is algebraic of degree 2. On each primary line, $H(t) = t^2 \sqrt{Q_L(t)}$ is determined by three seed invariants (κ, α, S_4) . The inverse limit exists and is automatically Maurer–Cartan because $D_{\mathcal{A}}^2 = 0$ (MC4).

In three dimensions, $\Theta_{\mathcal{A}}$ extends to an open/closed Maurer–Cartan element $\Theta_{\mathcal{A}}^{\text{oc}} = \Theta_{\mathcal{A}} + \sum_j \mu^{M_j}$ packaging the holomorphic-topological QFT on $\mathbb{C}_z \times \mathbb{R}_t$: the bar complex $B(\mathcal{A})$, coassociative over $(\text{ChirAss})^!$, supplies the holomorphic factorization data; the derived center pair $(\mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A}), \mathcal{A})$ carries the $\text{SC}^{\text{ch}, \text{top}}$ -algebra structure encoding both closed and open colours; and $\mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A})$ is the universal bulk (Volume II).

The bar complex thus organises five mathematical structures as projections of one object: the genus expansion $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ (UNIFORM-WEIGHT) is a GUE matrix model at $N^2 = \kappa$; the ordered bar $\bar{B}^{\text{ord}}(\mathcal{A})$ produces $U_q(\mathfrak{g})$ via FRT reconstruction; the quantum-group R -matrix yields knot invariants; the shadow metric Q_L defines a Bridgeland stability condition with universal wall phase $\pi/4$; and the Stokes constants $S_n/S_1 = (-1)^{n+1}n$ encode the resurgent structure.

Corollary (MC replaces the sum over topologies). On the Koszul locus, the tree-level complex carries a contracting homotopy (from the PBW degeneration), so $\Theta^{(g)}$ is determined up to gauge-equivalent MC elements by induction from $\Theta^{(0)}$. No free parameters appear at any genus; distinct choices of contracting homotopy yield gauge-equivalent MC elements; no choice of bulk topologies is required. The genus expansion is computed algebraically from the genus-0 chiral bracket by the clutching recursion of §4.6, not approximated perturbatively. The Maloney–Witten problem (negative Fourier coefficients in the Poincaré series) does not arise: the MC equation determines each F_g uniquely by algebraic induction, without summing over bulk topologies at fixed genus. The non-perturbative sum over distinct three-manifold topologies lies outside the MC framework.

5. The modular tangent complex and Chern–Weil theory

The MC element $\Theta_{\mathcal{A}}$ exists unconditionally and satisfies the MC equation. Its deformation theory is controlled by the modular tangent complex, obtained by twisting the convolution algebra by $\Theta_{\mathcal{A}}$ itself.

5.1. The L_{∞} -twisted differential

The MC element twists the convolution differential. For $x \in \mathfrak{g}_{\mathcal{A}}^{\text{mod}}$,

$$d_{\Theta_{\mathcal{A}}}(x) := \sum_{n \geq 0} \sum_{g \geq 0} \frac{\hbar^g}{n!} \ell_{n+1}^{(g)}(\Theta_{\mathcal{A}}^{\otimes n}, x) = d(x) + [\Theta_{\mathcal{A}}, x] + \frac{1}{2!} \ell_3^{(0)}(\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}, x) + \hbar \ell_1^{(1)}(x) + \dots$$

In the strict chart this collapses to $d_{\Theta_{\mathcal{A}}}(x) = d(x) + [\Theta_{\mathcal{A}}, x]$. The identity $d_{\Theta_{\mathcal{A}}}^2 = 0$ is the MC equation for $\Theta_{\mathcal{A}}$.

5.2. The modular tangent complex

The *modular tangent complex* at $\Theta_{\mathcal{A}}$:

$$T_{\Theta_{\mathcal{A}}}^{\text{mod}} := (\mathfrak{g}_{\mathcal{A}}^{\text{mod}}, d_{\Theta_{\mathcal{A}}}).$$

Its cohomology governs deformations: H^0 infinitesimal automorphisms (stabiliser), H^1 first-order deformations (tangent space), H^2 obstructions. $H^2 = 0$ makes the MC moduli smooth at $\Theta_{\mathcal{A}}$; $H^1 = 0$ makes $\Theta_{\mathcal{A}}$ rigid.

For a one-parameter family $\mathcal{A}_t$ varying in $k(t)$ or $c(t)$, the tangent complex forms a local system over the parameter space and the obstruction class traces a path in H^2 . The vanishing locus is the moduli of liftable deformations.

5.3. Three characteristic projections

Three successive projections of $\Theta_{\mathcal{A}}$ capture progressively finer structure: $\kappa(\mathcal{A})$ (leading scalar curvature), $\Delta_{\mathcal{A}}(x)$ (characteristic polynomial of a finite-rank operator), $\mathfrak{R}_4^{\text{mod}}(\mathcal{A})$ (first nonlinear characteristic class).

The scalar curvature.

$$\kappa(\mathcal{A}) = \text{tr}(\Theta_{\mathcal{A}})_{2,0} = \text{tr}(\pi \circ m_2 \circ \iota^{\otimes 2} \circ \Delta_{\text{sep}}) \in \mathbb{k}.$$

This is the first Chern class of the modular bar bundle: $\text{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ (UNIFORM-WEIGHT); multi-weight at $g \geq 2$ adds $\delta F_g^{\text{cross}}$. The Mumford relation $\lambda_g^{g+1} = 0$ confines genus- g scalar obstructions to dimension at most g .

Explicit values across the standard landscape:

Algebra $\mathcal{A}$	$\kappa(\mathcal{A})$	$\kappa(\mathcal{A}^!)$	Sum rule
Heisenberg $\mathcal{H}_k$	k	$-k$	0
Affine $\widehat{\mathfrak{g}}_k$	$\frac{(k+h^\vee) \dim \mathfrak{g}}{2h^\vee}$	$-\frac{(k+h^\vee) \dim \mathfrak{g}}{2h^\vee}$	0
Virasoro Vir_c	$c/2$	$(26-c)/2$	13
$\beta\gamma$ system	1	-1	0
Lattice V_Λ	$\text{rank}(\Lambda)$	$-\text{rank}(\Lambda)$	0
$\mathcal{W}_N(\widehat{\mathfrak{sl}}_N)$	$c \cdot (H_N - 1)$	$c' \cdot (H_N - 1)$	$K_N \cdot (H_N - 1)$
Free fermion ψ	1/4	-1/4	0

$H_N = \sum_{j=1}^N 1/j$ is the harmonic number. The sum rule $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = 0$ holds for Kac–Moody, free fields, lattice algebras. For $\mathcal{W}$ -algebras the sum is a nonzero constant: $\kappa(\text{Vir}_c) + \kappa(\text{Vir}_{26-c}) = 13$ and $\kappa(\mathcal{W}_{N,c}) + \kappa(\mathcal{W}_{N,c'}) = K_N(H_N - 1)$ with $K_N = c + c'$. Self-duality occurs at $c = 13$ for Virasoro and $c = 50$ for $\mathcal{W}_3$.

The spectral discriminant. The Koszul dual Hilbert series

$$h_{\mathcal{A}^!}(q) = \sum_{n \geq 0} \dim H^n(\bar{B}(\mathcal{A})) q^n$$

is rational for every standard family: it satisfies a holonomic recurrence of finite order d , and the transfer matrix T_{rec} is a $d \times d$ matrix. Define the *spectral branch object* as a finite-rank perfect complex $V_{\mathcal{A}}^{\text{br}}$ with *branch transport operator* $T_{\text{br},\mathcal{A}}: V_{\mathcal{A}}^{\text{br}} \rightarrow V_{\mathcal{A}}^{\text{br}}$ and

$$\Delta_{\mathcal{A}}(x) := \text{sdet}(1 - xT_{\text{br},\mathcal{A}}). \quad (0.0.13)$$

The rank of $V_{\mathcal{A}}^{\text{br}}$ equals the degree of algebraicity of $h_{\mathcal{A}^\dagger}$. Three properties: $\Delta_{\mathcal{A}}$ is self-dual ($\Delta_{\mathcal{A}^\dagger} = \Delta_{\mathcal{A}}$), unlike κ which is anti-symmetric for KM/free; exact factorization functors preserving quadratic OPE data preserve T_{br} ; the spectral radius equals $\dim \mathfrak{g}$ for affine KM at generic level.

For $\mathfrak{sl}_3$: the degree-3 recurrence has transfer matrix eigenvalues 8, $(3 \pm \sqrt{13})/2$, and $\Delta_{\widehat{\mathfrak{sl}_3}}(x) = (1-8x)(1-3x-x^2)$. For Heisenberg: $h_{\mathcal{H}_k^\dagger}(q) = 1+q$ and $\Delta_{\mathcal{H}_k}(x) = 1$. For $\beta\gamma$: $h = \sqrt{(1+q)/(1-3q)}$ and $\Delta_{\beta\gamma}(x) = (1-3x)(1+x)$.

In the Chern–Weil dictionary, $\Delta_{\mathcal{A}}$ plays the role of the Chern character: $\text{ch}_r(\mathcal{A}) = \frac{1}{r!} \frac{d^r}{dt^r} \big|_{t=0} [-\log \Delta_{\mathcal{A}}(t)] = \frac{1}{r} \text{tr}(T_{\text{br}}^r)$.

The quartic shadow.

$$\mathfrak{R}_{4,g,n}^{\text{mod}}(\mathcal{A}) = \text{pr}_{4,g,n}(\Theta_{\mathcal{A}}) \in H^*(\overline{\mathcal{M}}_{g,n}) \otimes (\bar{\mathcal{A}}^\vee)_{\text{cyc}}^{\otimes n}.$$

This is the first *nonlinear* characteristic class: the first invariant not recoverable from κ and $\Delta_{\mathcal{A}}$. Its genus-zero, four-point specialisation is the quartic contact invariant

$$Q_{\text{Vir}}^{\text{contact}} = \frac{10}{c(5c+22)}, \quad Q_{\mathcal{H}}^{\text{contact}} = 0, \quad Q_{\beta\gamma}^{\text{contact}} \neq 0.$$

For Heisenberg the quartic vanishes (no cubic OPE pole). For $\beta\gamma$ the quartic is the first gauge-nontrivial shadow (the cubic is gauge-removable). For Virasoro, both cubic and quartic are nonvanishing.

5.4. The Chern–Weil dictionary

The analogy between $\Theta_{\mathcal{A}}$ and a connection 1-form is exact. The modular bar bundle $B_{\text{mod}}^{\log}(\mathcal{A}) \rightarrow \overline{\mathcal{M}}_{g,n}$ is a principal bundle for the gauge group of the convolution algebra; $\Theta_{\mathcal{A}}$ is its flat connection (flatness automatic); the shadow tower is its Chern–Weil theory. Holonomy becomes factorisation homology, gauge equivalence becomes MC gauge, and the Weil homomorphism becomes the shadow algebra $\mathcal{A}^{\text{sh}} = H_\bullet(\text{Def}_{\text{cyc}}^{\text{mod}})$.

Classical Chern–Weil	Modular factorization
Principal G -bundle $P \rightarrow M$	Modular bar bundle $B_{\text{mod}}^{\log}(\mathcal{A}) \rightarrow \overline{\mathcal{M}}_{g,n}$
Connection $\omega \in \Omega^1(P, \mathfrak{g})$	MC element $\Theta_{\mathcal{A}} \in \text{MC}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}})$
Curvature $\Omega = d\omega + \frac{1}{2}[\omega, \omega]$	$D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$
Flatness $\Omega = 0$	Automatic (bar-intrinsic)
First Chern c_1	$\kappa(\mathcal{A}) = \text{tr}(\Theta_{\mathcal{A}})_{2,0}$
Chern character	Spectral determinant $\Delta_{\mathcal{A}}(t)$
Higher Chern classes c_r	Shadows $\mathfrak{R}_{4,g,n}^{\text{mod}}, \text{Sh}_r$ ($r \geq 5$)
Weil homomorphism	Shadow algebra $\mathcal{A}^{\text{sh}} = H_{\bullet}(\text{Def}_{\text{cyc}}^{\text{mod}})$
ad-invariant polynomial	Cyclic cocycle in $\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A})$
Cyclic pairing $\text{tr}(XY)$	Verdier duality on $\text{Ran}(X)$
Group multiplication	Operadic composition in $\mathcal{P}^{\text{ch}}$
Coproduct on $\mathcal{O}(G)$	Operadic cocomposition / clutching
Lagrangian splitting $T^*G = \mathfrak{g} \oplus \mathfrak{g}^*$	Complementarity $Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!)$
Holonomy $\text{Hol}(\gamma)$	Factorisation homology $\int_{\Sigma} \mathcal{A}$
Gauge equivalence	MC gauge in $\text{MC}_{\bullet}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}})$
Transgression $\text{CS}(\omega)$	Shadow obstruction tower (§6)
Flat sections $\nabla s = 0$	Twisted cochains $d_{\Theta_{\mathcal{A}}}(x) = 0$

5.5. The genus spectral sequence

The genus filtration $F^g L_{\text{mod}} := \bigoplus_{g' \geq g} (\mathfrak{g}_{\mathcal{A}}^{\text{mod}})_{g',*,*}$ yields a convergent spectral sequence. Write $D_0 = D_{\text{int}} + D_{\text{sep}}$ for the genus-preserving part and $D_1 = D_{\text{nsep}}$ for the genus-raising part; these satisfy $D_0^2 = 0$, $\{D_0, D_1\} = 0$, $D_1^2 = 0$ (a bicomplex).

The spectral sequence: $E_0^{p,q} = \text{gr}_F^p L_{\text{mod}}^{p+q}$ with $d_0 = [D_0, -]$; $E_1^{p,*}$ is the associated graded cohomology, genus by genus; the differential d_r at page r encodes the obstruction Ob_r ; $E_{\infty}^{p,q} \cong \text{gr}_F^p H^{p+q}(L_{\text{mod}})$. Convergence follows from completeness of the depth filtration and pronilpotence of the weight filtration; the $\varprojlim^1$ term vanishes by Mittag-Leffler.

This spectral sequence is distinct from the PBW spectral sequence (which decomposes by conformal weight within fixed genus). PBW computes *what* bar cohomology is; the genus spectral sequence determines *whether* the MC

element extends across genera. For Heisenberg it degenerates at E_1 . For Virasoro, $d_1 \neq 0$ and nontrivial differentials persist at every page.

5.6. Homotopy invariance and functoriality

The shadow algebra $\mathcal{A}^{\text{sh}} = H_\bullet(\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A}))$ is a homotopy invariant. For any ∞_α -quasi-isomorphism $f: \mathcal{A} \xrightarrow{\sim} \mathcal{A}'$,

$$f_*: \text{MC}_\bullet(\mathfrak{g}_{\mathcal{A}}^{\text{mod}}) \xrightarrow{\sim} \text{MC}_\bullet(\mathfrak{g}_{\mathcal{A}'}^{\text{mod}})$$

is a weak homotopy equivalence. The scalar κ , the spectral determinant $\Delta_{\mathcal{A}}$, and the full shadow obstruction tower $(\kappa, \Delta, \mathfrak{C}, \mathfrak{Q}, \text{Sh}_5, \dots)$ consist of quasi-isomorphism invariants; $\Theta_{\mathcal{A}}$ itself is a quasi-isomorphism invariant up to gauge. The assignment $\mathcal{A} \mapsto (\mathfrak{g}_{\mathcal{A}}^{\text{mod}}, \Theta_{\mathcal{A}})$ is a functor from chiral algebras with ∞_α -quasi-isos inverted to MC moduli problems. The convolution L_∞ -algebra is functorial in each slot separately but not as a bifunctor; functoriality of $\mathcal{A} \mapsto \mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ proceeds through the source slot.

5.7. The shadow CohFT and topological recursion

Projected to $\overline{\mathcal{M}}_{g,n}$, the MC equation produces tautological classes and tautological relations. The MC element determines shadow tautological classes $\tau_{g,n}(\mathcal{A}) \in R^*(\overline{\mathcal{M}}_{g,n+1})$ assembling into a cohomological field theory (Theorem ??): the factorisation axiom holds by the clutching law, the Σ_n -equivariance by the symmetry of stable-graph summation, the flat identity conditionally on the vacuum lying in V .

Projecting the MC equation to the (g, n) -component yields a tautological relation in $R^*(\overline{\mathcal{M}}_{g,n})$ at each (g, n) :

$$\sum_{\text{sep/nsep}} \xi_{\text{sep},*}(\tau_{g_1,|S|}(\mathcal{A}) \cdot \tau_{g_2,|S^c|}(\mathcal{A})) + \delta_{\text{pf}}^{(g,n)} = 0.$$

WDVV and Mumford from MC. At genus 0, degree 3: the MC tautological relation on $\overline{\mathcal{M}}_{0,4}$ reduces to WDVV. At degree 2, genus $g \geq 2$: substituting $\tau_{g,2}(\mathcal{A}) = \kappa \cdot \pi^* \lambda_g$ (UNIFORM-WEIGHT) recovers the Mumford λ -class clutching relation on $\overline{\mathcal{M}}_g$.

The complementarity propagator $P_{\mathcal{A}}: H^*(\mathcal{A}) \otimes H^*(\mathcal{A}^!) \rightarrow \mathbb{C}$ is the Givental R -matrix of the shadow CohFT; when it carries a flat unit (all standard families with vacuum in V), Teleman reconstruction recovers $\{\tau_{g,n}\}$ at all (g, n) from genus-0 data and $P_{\mathcal{A}}$. Eynard–Orantin topological recursion is the MC equation applied to the shadow obstruction tower.

MC datum	Tautological output
$\Theta_{\mathcal{A}} \in \text{MC}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}})$	shadow CohFT $\Omega_{g,n}^{\mathcal{A}}$
$D^2 = 0$ on FM^{log}	CohFT axioms
$D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ at (g, n)	tautological relation
$[\Theta, \Theta] _{\text{sep}}$	WDVV / associativity
$\hbar\Delta(\Theta) _{\text{nsep}}$	Mumford λ -class relations
$P_{\mathcal{A}}$	Givental R -matrix
MC recursion	EO topological recursion

The Pixton shadow bridge. For class-M algebras, $\Theta_{\mathcal{A}}$ generates tautological classes at every genus. At genus 2–3, the relations $\tau_{g,n}(\text{Vir}_c)$ for generic c produce elements of the Pixton ideal in $R^*(\overline{\mathcal{M}}_{g,n})$: Pixton’s tautological ring relations arise as shadows of the Virasoro MC equation. On the semisimple locus, MC-descended relations together with Mumford relations generate the Pixton ideal (Theorem ??); the non-semisimple frontier remains open.

The Böcherer bridge. At genus 2, the shadow tautological class $\tau_{2,0}(\text{Vir}_c)$ for lattice vertex algebras computes a central L -value via the Böcherer conjecture (Furusawa–Morimoto). For the Leech lattice, $\chi_{12} = \text{SK}(f_{22})$, and the MC-derived second Fourier coefficient $c_2 \approx -1.918 \times 10^{-6} \neq 0$ is the first nonzero central L -value produced by the shadow obstruction tower.

5.8. The primitive kernel and cofree reduction

$\Theta_{\mathcal{A}}$ lives in the cofree conilpotent coalgebra underlying the modular bar construction. By Milnor–Moore for modular operads, $\Theta_{\mathcal{A}}$ is determined by its restriction to primitive corollas and rigid cutting strata, the *primitive logarithmic modular kernel*

$$\mathfrak{K}_{\mathcal{A}} = \sum_{g \geq 0, n \geq 2} K_{g,n}^{\mathcal{A}} + \sum_{\rho \in \text{Rig}} K_{\rho}^{\mathcal{A}},$$

and the equivalence (Theorem ??)

$$(Q_{\mathcal{A}}^{\text{mod}})^2 = 0 \iff d\mathfrak{K}_{\mathcal{A}} + \mathfrak{K}_{\mathcal{A}} \star \mathfrak{K}_{\mathcal{A}} + \hbar\Delta_{\text{ns}}(\mathfrak{K}_{\mathcal{A}}) = 0,$$

with $\star$ the primitive pre-Lie product from log-FM residues, replaces the genus-by-genus obstruction programme by a single equation on primitive data.

Shell equations. Projecting to genus 2 and 3:

$$R_2(\mathfrak{K}_{\mathcal{A}}) = \Delta_{\text{ns}}(K_1) + \frac{1}{2}[K_1, K_1] + \sum_{\rho \in \text{Rig}_2} R_{\rho}^{\mathcal{A}}(K_{0,2}^{\mathcal{A}}),$$

$$R_3(\mathfrak{K}_{\mathcal{A}}) = \Delta_{\text{ns}}(K_2) + [K_1, K_2] + \frac{1}{3!}\ell_3(K_1, K_1, K_1) + \sum_{\rho} R_{\rho}^{\mathcal{A}}(\dots).$$

For Heisenberg, only Δ_{ns} survives. For affine Kac–Moody, $[K_1, K_1]$ contributes the cubic shadow. For $\beta\gamma$, cubic gauge triviality kills the bracket and the rigid stratum contributes the quartic contact invariant. For Virasoro and $\mathcal{W}_N$, all three terms are active at every genus.

The branch BV action. The finite-rank branch-active quotient $V_{\mathcal{A}}^{\text{br}} = V_{\mathcal{A}}^+ \oplus V_{\mathcal{A}}^-$ carries a BV master action

$$S_{\mathcal{A}}^{\text{br}}(x, \xi) = \frac{1}{2}\Omega_{\mathcal{A}}(K_{\mathcal{A}}^{\text{br}}x, x) + \sum_{m+n \geq 3} \frac{1}{m!n!} \mu_{m,n}^{\mathcal{A}}(x^{\otimes m}; \xi^{\otimes n}),$$

satisfying $\hbar \Delta_{\mathcal{A}}^{\text{br}} S_{\mathcal{A}}^{\text{br}} + \frac{1}{2}\{S_{\mathcal{A}}^{\text{br}}, S_{\mathcal{A}}^{\text{br}}\}_{\text{BV}} = 0$. The metaplectic half-density $\delta_{\mathcal{A}} \in 1 + x\mathbb{k}[[x]]$ satisfies $\delta_{\mathcal{A}}^2 = \Delta_{\mathcal{A}} = \det(1 - xT_{\mathcal{A}}^{\text{br,red}})$: the spectral discriminant is the square of a canonical half-density.

5.9. Entanglement from the modular characteristic

The bar coalgebra carries a natural bipartite structure: its coproduct splits a configuration of points into two halves, one on each side of a spatial cut. For a spatial bipartition of an interval of length L with UV cutoff ε , the scalar projection of $\Theta_{\mathcal{A}}$ restricted to the bipartite cut yields

$$S_{\text{EE}}(\mathcal{A}) = \frac{c(\mathcal{A})}{3} \log \frac{L}{\varepsilon}.$$

For Virasoro $c = 2\kappa$; the relationship between c and κ is family-dependent. Complementarity gives the entanglement sum rule $S_{\text{EE}}(\mathcal{A}) + S_{\text{EE}}(\mathcal{A}^!) = \frac{c(\mathcal{A}) + c(\mathcal{A}^!)}{3} \log(L/\varepsilon)$; for the Virasoro family $\text{Vir}_c^! = \text{Vir}_{26-c}$ this gives $c + c' = 26$ and the universal sum $26/3 \cdot \log(L/\varepsilon)$.

The four shadow-depth classes (G/L/C/M) classify entanglement complexity: class G (Gaussian, $r_{\text{max}} = 2$) exhibits area-law entanglement only; class L ($r_{\text{max}} = 3$) adds subleading logarithmic corrections from the cubic shadow; class C ($r_{\text{max}} = 4$) adds a contact correction from the quartic invariant; class M ($r_{\text{max}} = \infty$) carries an infinite tower of corrections.

The entanglement programme (G11–G16). Six theorem targets extend the scalar result. G11: Ryu–Takayanagi from the scalar projection of $\Theta_{\mathcal{A}}$. G12: Koszulness as exact quantum error correction; the Knill–Laflamme condition is the Lagrangian isotropy condition from the complementarity pairing; shadow depth equals the number of redundancy channels; the twelve Koszulness characterisations translate into a twelve-fold code dictionary; failure of Koszulness is failure of the code. G13: replica trick from the MC equation. G14: modular flow from the tangent complex. G15: QES as the shadow connection Ward identity. G16: Page curve at $c = 13$ from the shadow connection at the self-dual point; the scrambling time is controlled by $\Delta_{\mathcal{A}}$.

6. The shadow obstruction tower

6.1. The extension tower

Can $\Theta_{\mathcal{A}}$ be approximated by finitely many degrees, and what controls the passage from one finite approximation to the next? The total weight filtration answers both.

Define the *total weight* $w(g, r, d) := 2g - 2 + r + d$. The filtration $F^N \mathfrak{g}_{\mathcal{A}}^{\text{amb}} := \bigoplus_{w \geq N} (\mathfrak{g}_{\mathcal{A}}^{\text{amb}})_{g, r, d}$ is exhaustive, separated, complete. $\mathfrak{g}_{\mathcal{A}}^{\text{amb}}$ is the *ambient* (planted-forest-extended) modular convolution algebra. The *modular extension tower*:

$$\mathcal{E}_{\mathcal{A}}(N) := \text{MC}(\mathfrak{g}_{\mathcal{A}}^{\text{amb}}/F^{N+1}), \quad N \geq 1,$$

with restriction maps $\mathcal{E}_{\mathcal{A}}(N+1) \rightarrow \mathcal{E}_{\mathcal{A}}(N)$. A full lift is a compatible point of the inverse system; the bar-intrinsic construction provides one unconditionally. The shadow obstruction tower studies $\mathcal{E}_{\mathcal{A}}(1) \leftarrow \mathcal{E}_{\mathcal{A}}(2) \leftarrow \mathcal{E}_{\mathcal{A}}(3) \leftarrow \cdots$.

6.2. Obstruction classes and the all-degree master equation

Define $\Theta_{\mathcal{A}}^{\leq r}$ by projecting to total weight $\leq r$. The *shadow obstruction tower* is the sequence $\Theta_{\mathcal{A}}^{\leq 2}, \Theta_{\mathcal{A}}^{\leq 3}, \Theta_{\mathcal{A}}^{\leq 4}, \dots$. At each truncation the MC equation fails by a remainder,

$$D\Theta_{\mathcal{A}}^{\leq r} + \frac{1}{2}[\Theta_{\mathcal{A}}^{\leq r}, \Theta_{\mathcal{A}}^{\leq r}] = -o_{r+1}(\mathcal{A}),$$

with obstruction class $o_{r+1}(\mathcal{A}) \in Z^2(F^{r+1}/F^{r+2}, d_{\leq r})$. Extension is possible iff $[o_{r+1}] = 0$ in $H^2(F^{r+1}/F^{r+2}, d_{\leq r})$. The bar-intrinsic construction guarantees extension, so the obstruction class is always exact; the shadow obstruction tower records the gauge choices at each stage.

The all-degree master equation at degree r :

$$\nabla_H(\text{Sh}_r(\mathcal{A})) + o^{(r)}(\mathcal{A}) = 0, \quad r \geq 3,$$

with ∇_H the Hodge covariant derivative. This equation determines Sh_r uniquely modulo gauge if $H^1(F^r/F^{r+1}, d_{\leq r-1}) = 0$.

6.3. Cubic gauge triviality and the canonical quartic

Cubic gauge triviality. When $H^1(F^3/F^4, d_2) = 0$, the cubic MC term $\Theta_{\mathcal{A}}^{[3]}$ is gauge-trivial: there exists $\exp(\xi_3)$ with $\xi_3 \in F^3$ such that $\exp(\xi_3) \cdot \Theta_{\mathcal{A}}^{\leq 3}$ has vanishing weight-3 component.

After this gauge transformation, the quartic obstruction class $[\mathfrak{Q}(\mathcal{A})] \in H^2(F^4/F^5, d_2)$ is canonical: independent of homotopy retraction data, cubic gauge, and MC gauge orbit representative. This is the first characteristic class not polynomial in κ . The mechanism is standard Postnikov: trivial H^1 at level n makes the H^2 class at level $n+1$ canonical.

The clutching law for the quartic follows from Mok's log FM degeneration formula. For $C \rightsquigarrow C_0 = C_1 \cup_p C_2$,

$$\mathfrak{Q}(\mathcal{A})|_{C_0} = \sum_{a+b=4} \mathfrak{Q}^{(a)}(\mathcal{A})|_{C_1} \cdot \mathfrak{Q}^{(b)}(\mathcal{A})|_{C_2} + \text{planted-forest correction}.$$

The correction is genuinely logarithmic, with no DM counterpart.

The quartic contact invariant $Q_{\text{vir}}^{\text{contact}} = 10/[c(5c+22)] \in H^2(F^4/F^5, d_2)$. The denominator has physical meaning: $c=0$ is the logarithmic threshold; $c=-22/5$ is the minimal model accumulation point $\mathcal{M}(2,5)$. The numerator $10 = \binom{5}{2}$ counts degree-four Catalan channels.

6.4. The Hamilton–Jacobi generating function

For a multi-generator algebra $(\mathcal{W}_3$ with generators T (weight 2) and W (weight 3)), the deformation space is two-dimensional. The Riccati ODE lifts to a Hamilton–Jacobi PDE. Write $U = \sum_{s \geq 3} \text{Sh}_s(x_T, x_W)$; the master equation becomes

$$2E(U) + \frac{1}{2} \|\nabla U\|_H^2 = R(x_T, x_W),$$

with Euler operator $E = x_T \partial_T + x_W \partial_W$, kinetic energy $\|\nabla U\|_H^2 = (2/c)(\partial_T U)^2 + (3/c)(\partial_W U)^2$, and R a polynomial of degree ≤ 4 in $(\kappa, \mathfrak{C}, \mathfrak{Q})$. The gradient $dU: X \rightarrow T^*X$ embeds the deformation space as a Lagrangian submanifold

of T^*X . Whether the genus expansion admits a quantisation with U as leading semiclassical term is open: it would require a quantum Hamiltonian whose WKB expansion has leading term U and whose quantum corrections reproduce the genus- g shadow amplitudes.

On the T -line the conformal weight forces the transverse momentum to vanish and the PDE collapses to the Riccati ODE. On the W -line the cubic shadow $\text{Sh}_3 = 2x_T^3 + 3x_T x_W^2$ has nonzero normal derivative $(\partial_T \text{Sh}_3)|_{x_T=0} = 3x_W^2$, and the first inter-channel coupling correction enters at degree 6:

$$\delta_6^{W\text{-line}} = \frac{576(5c + 38)}{c^3(5c + 22)^3}.$$

The T -line is autonomous; the W -line is not. The asymmetry is intrinsic to the weight difference between T (weight 2) and W (weight 3).

6.5. The four depth classes

The *shadow depth* $r_{\max}(\mathcal{A})$ is the smallest r such that $o_{r+1}(\mathcal{A}) = 0$ for all subsequent levels, or ∞ if the tower never terminates. The standard landscape partitions into four depth classes; no other values occur on any one-dimensional primary slice. The single-line dichotomy: the shadow metric $Q_L(t)$ is quadratic in (κ, α, S_4) , and its factorisation type (perfect square or irreducible) determines whether the tower terminates.

Class	Symbol	$r_{\max}$	$\ell_3^{(0)}, \ell_4^{(0)}$	Potential	Examples
Gaussian	G	2	0, 0	quadratic	Heisenberg $\mathcal{H}_k$, free fermion ψ
Lie/tree	L	3	$\neq 0, 0$	cubic	Affine $\widehat{\mathfrak{sl}}_N$, currents
Contact	C	4	$0, \neq 0$	quartic	$\beta\gamma$, bc ghosts
Mixed	M	∞	$\neq 0, \neq 0$	non-polynomial	Virasoro, $\mathcal{W}_N$, $\mathcal{W}_\infty$

Gaussian ($r_{\max} = 2$): transferred brackets $\ell_k^{(0)}$ vanish for $k \geq 3$ on bar cohomology, so the tower terminates at κ . Two mechanisms: Heisenberg has no simple pole; the free fermion has a simple pole but fermionic antisymmetry forces $S_3 = 0$. Sewing is by Fredholm determinant (Heisenberg) or Pfaffian (fermion). **Lie/tree** ($r_{\max} = 3$): the OPE has a simple pole generating a Lie bracket, but $\ell_4^{(0)}$ vanishes on bar cohomology. **Contact** ($r_{\max} = 4$): the simple-pole bracket vanishes on bar cohomology, but the quaternary bracket survives; the first nontrivial shadow is $\mathfrak{Q}(\mathcal{A})$. **Mixed** ($r_{\max} = \infty$): both

cubic and quartic shadows are nontrivial, and their Poisson bracket $\{\mathfrak{C}, \mathfrak{Q}\}_H$ generates a nonvanishing quintic, then sextic, and so on.

Shadow depth is not Koszulness. All four depth classes contain chirally Koszul algebras. Heisenberg (depth 2), affine at generic level (depth 3), $\beta\gamma$ (depth 4), and Virasoro (depth ∞) are all chirally Koszul. Shadow depth classifies the complexity of Koszul algebras, not Koszulness itself.

The *operadic complexity theorem* (Theorem ??):

$$r_{\max}(\mathcal{A}) = A_{\infty}\text{-depth}(\mathcal{A}) = L_{\infty}\text{-formality level}(\mathcal{A}).$$

A_{∞} -depth is the largest r with $m_r \neq 0$ in the minimal A_{∞} model; L_{∞} -formality level is the largest r with a nontrivial r -ary bracket on bar cohomology not removable by L_{∞} -isomorphism.

6.6. The genus–degree bigrading

The MC equation decomposes along two orthogonal axes. The *genus axis* extends $\Theta_{\mathcal{A}}$ from genus 0 to 1 to 2, controlled by the genus spectral sequence. The *degree axis* extends $\Theta_{\mathcal{A}}^{\leq 2} \rightarrow \Theta_{\mathcal{A}}^{\leq 3} \rightarrow \Theta_{\mathcal{A}}^{\leq 4}$, controlled by cyclic obstruction classes. At fixed degree 2 the genus tower is governed by κ alone (UNIFORM-WEIGHT); at fixed genus 0 each new shadow coefficient introduces independent data. The axes couple through the visibility formula $g_{\min}(S_r) = \lfloor r/2 \rfloor + 1$: S_r first contributes at genus $\lfloor r/2 \rfloor + 1$ and two new shadow coefficients enter at each genus.

6.7. Genus-two shells

Genus 2 is the first level where all three boundary strata interact:

$$\Theta_{\mathcal{A}}^{(2)} = \underbrace{\Theta_{\text{looploop}}^{(2)}}_{\text{iterated BV}} + \underbrace{\Theta_{\text{sepoloop}}^{(2)}}_{\text{clutching} \times \text{BV}} + \underbrace{\Theta_{\text{pf}}^{(2)}}_{\text{planted-forest}}.$$

The iterated-BV term involves the figure-eight graph:

$$\Theta_{\text{loop}^2}^{(2)} = \frac{1}{2} \text{tr}_1 \text{tr}_2 (P_{\mathcal{A}}^{\otimes 2} \circ m_{0,4}^{(h)} \circ \Delta^{\otimes 2}).$$

The separating-times-loop term:

$$\Theta_{\text{sepoloop}}^{(2)} = \text{tr}_{\text{ns}} (P_{\mathcal{A}} \circ \ell_2^{(0)}(\Theta^{(1,\bullet)}, \Theta^{(1,\bullet)})).$$

The planted-forest term is the genuinely logarithmic contribution, present only when cubic or quartic shadows are nontrivial.

Family-by-family structure:

	$\Theta_{\text{loop}^2}^{(2)}$	$\Theta_{\text{sepoloop}}^{(2)}$	$\Theta_{\text{pf}}^{(2)}$	Depth
Heisenberg $\mathcal{H}_k$	✓	—	—	G
Affine $\widehat{\mathfrak{g}}_k$	✓	✓	—	L
$\beta\gamma$ system	✓	—	—	C
Virasoro Vir_c	✓	✓	✓	M
$\mathcal{W}_N$	✓	✓	✓	M

Heisenberg: only iterated BV. Affine: adds separating-times-loop. Virasoro and $\mathcal{W}_N$: all three active, with planted-forest carrying the quartic resonance. $\beta\gamma$: only iterated BV survives; the separating term vanishes because the Lie bracket is trivial on bar cohomology, and the planted-forest vanishes because the quartic contact invariant $\mu_{\beta\gamma} = 0$ (rank-one abelian rigidity). Contact termination via the iterated BV shell, not Gaussian collapse.

6.8. The genus loop operator

The *genus loop operator* $\Lambda_P: \text{Sym}^r(\bar{\mathcal{A}}^\vee)_{\text{cyc}} \rightarrow \text{Sym}^{r-2}(\bar{\mathcal{A}}^\vee)_{\text{cyc}}$ contracts two legs with the propagator:

$$\Lambda_P(\phi)(a_1, \dots, a_{r-2}) = \sum_i \phi(a_1, \dots, a_{r-2}, e_i, e^i).$$

This raises genus by 1 and lowers degree by 2: the shadow-level incarnation of Δ_{ns} .

Explicit Virasoro computation. On the Virasoro primary line, the conformal-weight-2 bar cohomology is one-dimensional, spanned by the class x dual to T . The Killing form is $H = (c/2)x^2$, the propagator $P = 2/c$, and the quartic contact invariant is $Q_{\text{Vir}}^{\text{contact}} \cdot x^4$ with $Q_{\text{Vir}}^{\text{contact}} = 10/[c(5c+22)]$. Applying Λ_P :

$$\Lambda_P\left(\frac{10}{c(5c+22)} \cdot x^4\right) = \binom{4}{2} \cdot \frac{2}{c} \cdot \frac{10}{c(5c+22)} \cdot x^2 = \frac{120}{c^2(5c+22)} x^2,$$

the genus-one Hessian correction. The loop ratio

$$\rho_{\text{Vir}}^{(1)} = \frac{\Lambda_P(Q \cdot x^4)}{\kappa \cdot x^2} = \frac{120/[c^2(5c+22)]}{c/2} = \frac{240}{c^3(5c+22)}$$

is not determined by $\kappa = c/2$: it detects the quartic structure through its genus-one projection. Asymptotically $\rho^{(1)} \sim 48/c^4$ for $c \rightarrow \infty$.

6.9. Quintic forcing and infinite towers

The quintic obstruction determines whether the tower terminates at degree 4.

Virasoro. Both $\mathfrak{C}_{\text{Vir}}$ and $Q_{\text{Vir}}^{\text{contact}}$ are nonzero. Their Poisson bracket

$$o_{\text{Vir}}^{(5)} = \{\mathfrak{C}_{\text{Vir}}, Q_{\text{Vir}}^{\text{contact}}\}_H = \frac{480}{c^2(5c+22)}x^5 \neq 0.$$

For all $c \neq 0, -22/5$, the Virasoro shadow tower does not terminate; $r_{\max}(\text{Vir}_c) = \infty$. Same for every $\mathcal{W}_N$.

Heisenberg. $\mathfrak{C}_{\mathcal{H}} = 0$, $Q_{\mathcal{H}}^{\text{contact}} = 0$, hence $o^{(5)} = 0$ and all higher obstructions vanish. $r_{\max} = 2$.

Affine. $\mathfrak{C}_{\mathfrak{g}} \neq 0$ but $Q_{\mathfrak{g}}^{\text{contact}} = 0$ (quaternary bracket vanishes on bar cohomology by the PBW theorem). $o^{(5)} = 0$; $r_{\max} = 3$.

$\beta\gamma$. $\mathfrak{C}_{\beta\gamma} = 0$ (Lie bracket trivial on bar cohomology) but $Q_{\beta\gamma}^{\text{contact}} \neq 0$. $o^{(5)} = 0$; $r_{\max} = 4$.

	$\mathfrak{C}$	$\mathfrak{Q}$	$o^{(5)} = \{\mathfrak{C}, \mathfrak{Q}\}_H$	$r_{\max}$
Heisenberg	0	0	0	2
Affine	$\neq 0$	0	0	3
$\beta\gamma$	0	$\neq 0$	0	4
Virasoro	$\neq 0$	$\neq 0$	$\neq 0$	∞

Finite termination requires $\mathfrak{C} = 0$ or $\mathfrak{Q} = 0$; when both are nonzero, their bracket generates the quintic, and the induction continues without bound.

The shadow growth rate. For class M algebras, the Taylor coefficients S_r of $\sqrt{Q_L}$ grow as $S_r \sim A \cdot \rho(\mathcal{A})^r \cdot r^{-5/2} \cos(r\theta + \varphi)$ with shadow growth rate $\rho(\mathcal{A}) = \sqrt{9\alpha^2 + 2\Delta}/(2|\kappa|)$, a continuous invariant. Self-dual Virasoro ($c = 13$) has $\rho \approx 0.467$. The critical cubic $5c^3 + 22c^2 - 180c - 872 = 0$ has root $c^* \approx 6.125$: the tower converges for $c > c^*$ and diverges for $c < c^*$.

The shadow partition function. The double sum $Z_{\mathcal{A}}^{\text{sh}} = \sum_{g,r} \hbar^{2g-2} Z_g^{(r)}$ converges absolutely. Genus decay at rate $1/(2\pi)^{2g}$ (Faber–Pandharipande); degree decay at rate $\rho^r r^{-5/2}$ (Darboux analysis). The string partition function diverges as $(2g-2)!$. For Virasoro at $c = 26$, $\rho \approx 0.234$ and Z^{sh} has positive radius of convergence in both $\hbar$ and degree.

6.10. Independent sum factorisation

For a direct sum $L = L_1 \oplus L_2$ with vanishing mixed OPE, all shadow invariants separate:

$$\begin{aligned}\kappa(L_1 \oplus L_2) &= \kappa(L_1) + \kappa(L_2), \\ T^{\text{br}}(L_1 \oplus L_2) &= T^{\text{br}}(L_1) \oplus T^{\text{br}}(L_2), \\ \Delta_{L_1 \oplus L_2}(t) &= \Delta_{L_1}(t) \cdot \Delta_{L_2}(t), \\ R_4^{\text{mod}}(L_1 \oplus L_2) &= R_4^{\text{mod}}(L_1) + R_4^{\text{mod}}(L_2).\end{aligned}$$

The full MC element factorises: $\Theta_{L_1 \oplus L_2} = \Theta_{L_1} \hat{\otimes} 1 + 1 \hat{\otimes} \Theta_{L_2}$, and $Z_{L_1 \oplus L_2} = Z_{L_1} \cdot Z_{L_2}$.

Every algebra in the standard landscape is either primitive (Heisenberg, Virasoro, $\beta\gamma$) or built from primitive pieces (affine = current algebra over simple Lie, lattice = exponential extension of Heisenberg, $\mathcal{W}_N$ = quantum Drinfeld–Sokolov reduction of affine). Drinfeld–Sokolov reduction is the *shadow-depth escalator* (Principle ??): it sends current algebras (class L, $r_{\text{max}} = 3$, finite towers) to $\mathcal{W}$ -algebras (class M, $r_{\text{max}} = \infty$, infinite towers). The nonlinearity of the Dirac bracket on the Slodowy slice creates the quartic and all higher shadows.

6.11. The representation-theoretic perspective

The categorical logarithm $\eta_{ij} = d \log(z_i - z_j)$ of §1 is the integral kernel of a tautology: $D_{\mathcal{A}}^2 = 0$ on the modular bar complex. Three structure theorems emerge. *Algebraicity*: on each primary slice L , $H(t)^2 = t^4 Q_L(t)$ with Q_L quadratic in κ, α, S_4 . *Formality*: the genus-0 shadow obstruction tower is the L_∞ formality obstruction tower; the four depth classes G/L/C/M are the corresponding formality types, while positive-genus corrections form a separate quantum layer. *Complementarity*: for a Koszul pair $(\mathcal{A}, \mathcal{A}^\dagger)$, the summands $\mathbf{Q}_g(\mathcal{A}) \oplus \mathbf{Q}_g(\mathcal{A}^\dagger)$ are Lagrangian (Theorem C).

The discriminant $\Delta = 8\kappa S_4$ governs the single-line dichotomy: $\Delta = 0$ forces tower termination (L_∞ -formality); $\Delta \neq 0$ forces an infinite tower (intrinsic non-formality). The contact class C ($\beta\gamma$) escapes via stratum separation. No invariant of the classical bar complex of a graded algebra sees Δ : it is visible only on curves of positive genus, where the Arnold relation acquires a defect.

Classical Koszul duality (Priddy, Beilinson–Ginzburg–Soergel) embeds into the genus 0, degree 2, $\Delta = 0$ stratum via formal-disk restriction (the formal, class-G case); the E_1/E_∞ distinction and Fulton–MacPherson geometry are already absent over a bare point. The R -matrix of quantum

groups arises as the genus-0 shadow $\text{Sh}_{0,2}(\Theta_{\mathfrak{g}_k})$ (§7.7). At the critical level $k = -h^\vee$, the scalar class vanishes ($\kappa = 0$) and bar cohomology recovers the Feigin–Frenkel center $\text{Fun}(\text{Op}_{L_{\mathfrak{g}}})$; the Langlands programme lives on this orthogonal axis, not in the MC element but in the cohomology that emerges on the uncurved scalar locus.

9. Scalar saturation and the Koszulness programme

The bar construction works for all chiral algebras. It works best, the cobar inverts the bar, the spectral sequence collapses, the resolution is linear, precisely on the Koszul locus. Characterizing this locus is the central foundational question introduced in §3.3 and anticipated throughout sections 4 and 5. Two complementary results: scalar saturation (the MC element is determined by a single scalar for all standard families) and the Koszulness meta-theorem (twelve characterisations: of which eight (among (i)–(x)) are genuinely independent bidirectional equivalences, one ((vi)) is a consequence of (v), one ((viii)) is a one-way chiral Hochschild consequence on the Koszul locus, one ((xi)) is conditional on perfectness, and one ((xii)) is one-directional).

9.1. Scalar saturation

For a one-parameter algebraic family $\mathcal{A}_k$ with rational OPE coefficients (structure constants rational functions of k), the universal MC element satisfies

$$\Theta_{\mathcal{A}_k} = \kappa(\mathcal{A}_k) \cdot \eta_{\text{univ}} \otimes \Lambda_{\text{univ}} + (\text{terms in } F^3 \mathfrak{g}_{\mathcal{A}_k}^{\text{mod}}).$$

Scalar saturation universality: higher-order terms vanish. For every algebraic family with rational OPE coefficients, $\Theta_{\mathcal{A}_k}$ is determined by $\kappa(\mathcal{A}_k)$ modulo terms in the degree- ≥ 3 filtration.

Step 1: Whitehead reduction. For a vertex algebra with reductive symmetry at generic level, the primitive cyclic cohomology $H_{\text{cyc,prim}}^2(\mathfrak{g}^{\text{mod}}/F^3, d_2)$ is computed by a Whitehead-type spectral sequence with $E_2^{p,q} = H^p(\mathfrak{g}; \text{Sym}^q V)$. For reductive $\mathfrak{g}$ and generic level, $H^2(\mathfrak{g}; \text{Sym}^q V) = 0$ by the Whitehead lemma, and the degree-2 MC class lifts uniquely.

Step 2: Algebraic semicontinuity. The vanishing locus

$$E = \{k \in \mathbb{C} : H_{\text{cyc,prim}}^2(\mathcal{A}_k) \neq 0\}$$

is Zariski-closed by upper semicontinuity. At $k = \infty$ the algebra is Heisenberg and $H_{\text{cyc,prim}}^2 = 0$, so E is a proper Zariski-closed subset. Explicit computation verifies $E = \emptyset$ for $\mathfrak{sl}_N$ ($N = 2, \dots, 10$): no exceptional level. The result

covers the entire standard Lie-theoretic landscape at all non-critical levels, including admissible.

Scope. Scalar saturation applies to every family in the standard landscape: affine Kac–Moody (all types, all non-critical levels), Virasoro (all c), $\mathcal{W}_N$ (all N , all non-degenerate levels), $\beta\gamma$ and lattice algebras (all parameters). The residual conjecture, that $E = \emptyset$ for every positive-energy chiral algebra with rational OPE coefficients, is open only for non-algebraic-family constructions.

9.2. The Koszulness meta-theorem

The Koszul condition admits twelve formulations, spanning homological algebra (PBW collapse, Ext vanishing), homotopy theory (A_∞ formality, E_2 formality), operadic algebra (Barr–Beck–Lurie, FM acyclicity), modular geometry (FH concentration, tropical Koszulness), and symplectic topology (Lagrangian criterion). Ten are unconditionally equivalent; one (the Lagrangian criterion) is conditional on perfectness/nondegeneracy; one ($\mathcal{D}$ -module purity) is one-directional (forward direction proved, converse open).

For $\mathcal{A}$ positive-energy and finitely strongly generated, the following are equivalent:

- (i) $\mathcal{A}$ is chirally Koszul.
- (ii) The PBW spectral sequence on $\bar{B}^{(g)}(\mathcal{A})$ collapses at E_2 for every g .
- (iii) The minimal A_∞ -model of $H^\bullet(\bar{B}(\mathcal{A}))$ is formal: $m_n = 0$ for $n \geq 3$.
- (iv) $\text{Ext}_{\mathcal{A}}^{p,q}(\omega_X, \omega_X) = 0$ for $p \neq q$.
- (v) The bar-cobar counit $\Omega(\bar{B}(\mathcal{A})) \rightarrow \mathcal{A}$ is a quasi-isomorphism.
- (vi) The Barr–Beck–Lurie comparison for $\bar{B} \dashv \Omega$ is an equivalence.
- (vii) $\int_X \mathcal{A}$ is concentrated in degree 0 for every smooth proper X .
- (viii) $\text{ChirHoch}^*(\mathcal{A})$ vanishes outside degrees $\{0, 1, 2\}$.
- (ix) The Kac–Shapovalov determinant $\det G_h \neq 0$ in the bar-relevant range.
- (x) The restriction $i_S^! \bar{B}_n(\mathcal{A})$ to every FM boundary stratum S is acyclic outside degree 0.

Under the additional hypothesis that the ambient tangent complex is perfect and nondegenerate:

- (xi) $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^\dagger}$ are transverse Lagrangians in the (-1) -shifted symplectic deformation space $\mathcal{M}_{\text{comp}}$.

Finally (xii) $\Rightarrow$ (x), where:

- (xii) Each $\bar{B}_n(\mathcal{A})$ is pure of weight n as a mixed Hodge module, with characteristic variety aligned to FM boundary strata ($\mathcal{D}$ -module purity).

The converse of (xii) is open.

The proof circuit. The core equivalence chain is (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (v) $\Leftrightarrow$ (viii), proved by PBW concentration $\rightarrow$ bar-cobar inversion $\rightarrow$ chiral Hochschild polynomial growth $\rightarrow$ Koszul-complex acyclicity, with the A_∞ formality branch via HPL transfer and Keller classicality. From this core: (iv) by Ext spectral sequence, (vi) by conservativity with log-FM totalization, (vii) by $\bar{B} = \int_X$, (ix) by non-degeneracy forcing PBW injectivity, (x) by stratum-by-stratum PBW concentration. The Lagrangian criterion (xi) uses PTVV derived intersection theory on the perfectness hypothesis. The $\mathcal{D}$ -module purity implication (xii) $\Rightarrow$ (x) uses weight filtration on the bar complex; the converse would require concentration to force purity. The cycle closes via (viii) $\Rightarrow$ (i): Koszul-complex acyclicity implies the defining characterisation.

Beyond the meta-theorem. Six additional characterisations are proved in the text:

- (a) *Shadow-formality at low degree.* The shadow obstruction tower $\Theta_{\mathcal{A}}^{\leq r}$ equals the L_∞ -formality obstruction tower of $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ at degrees $r = 2, 3, 4$ (Proposition ??). Full equivalence to (iii) requires all degrees.
- (b) *E_2 -formality of chiral Hochschild cohomology.* Koszulness implies $\text{ChirHoch}^*(\mathcal{A})$ is formal as an E_2 -algebra (Deligne conjecture applied to the bar-cobar resolution).
- (c) *Genus-0 curve independence.* Koszulness on $\mathbb{P}^1$ implies Koszulness on every smooth proper curve (Proposition ??). Proved independently by factorization descent.
- (d) *PBW universality* (sufficient). Every freely strongly generated vertex algebra is chirally Koszul (Proposition ??). Applies to universal algebras: $V_k(\mathfrak{g})$ is Koszul at every level including critical and admissible. For simple quotients $L_k(\mathfrak{g})$ at admissible levels, status is rank-dependent: $L_k(\mathfrak{sl}_2)$ is Koszul at all admissible levels, but at $\text{rk}(\mathfrak{g}) \geq 2$ with denominator $q \geq 3$, Koszulness fails via a $\text{rk}(\mathfrak{g})$ -dimensional bar obstruction from the abelian Cartan (Theorem ?? at $\mathfrak{g} = \mathfrak{sl}_3$; Conjecture ?? at $\text{rk}(\mathfrak{g}) \geq 3$). PBW universality is one-directional: not every Koszul algebra is freely strongly generated.
- (e) *Tropical Koszulness.* $\mathcal{A}$ is chirally Koszul iff $\bar{B}^{\text{trop}}(\mathcal{A})$ is acyclic. Proved by the weight spectral sequence on the real blowup.
- (f) *Bifunctor obstruction decomposition* (one-directional). Koszulness implies the RNW one-slot obstruction decomposes into independent bar-side and cobar-side components (Theorem ??). The converse is open.

Status summary.

Meta-theorem (Theorem ??):

(i)–(v), (vii) _{g=0} , (ix), (x)	8 genuinely independent bidirectional	proved
(vi) Barr–Beck–Lurie	consequence of (v)	proved
(viii) chiral Hochschild $\{0, 1, 2\}$	one-way consequence on Koszul locus	proved
(xi) Lagrangian	conditional on perfectness	proved (uncondition)
(xii) $\mathcal{D}$ -module purity	(xii) $\Rightarrow$ (x) only	converse open

Additional characterisations:

(a) Shadow-formality	equivalence at degrees 2, 3, 4	proved
(b) E_2 -formality	consequence of Koszulness	proved
(c) Curve independence	independent theorem	proved
(d) PBW universality	sufficient condition	proved
(e) Tropical Koszulness	equivalence	proved
(f) Bifunctor decomposition	one-directional	converse open

9.3. Shadow depth and operadic complexity

The four depth classes $\mathbf{G}/\mathbf{L}/\mathbf{C}/\mathbf{M}$ admit refinement: in general $r_{\max} \in \{2, 3, 4, 5, \dots\} \cup \{\infty\}$, with class $\mathbf{F}_r$ (tower terminates at degree r) structurally permitted for all $r \geq 2$; $\mathbf{G} = \mathbf{F}_2$, $\mathbf{L} = \mathbf{F}_3$, $\mathbf{C} = \mathbf{F}_4$ name the standard-landscape representatives.

The operadic complexity theorem (Theorem ??):

$$r_{\max}(\mathcal{A}) = A_{\infty}\text{-depth}(\mathcal{A}) = L_{\infty}\text{-formality level}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}}).$$

A_{∞} -depth is the minimal r with $m_k = 0$ for $k > r$ on $H^{\bullet}(\bar{B}(\mathcal{A}))$; L_{∞} -formality level is the minimal r such that the L_{∞} -structure on $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ is formal through degree r . Three consequences: the shadow obstruction tower *is* the L_{∞} -formality obstruction tower; the depth classification becomes a structural invariant of the operadic type of $\mathcal{A}$; the tower of a Gaussian algebra is a quadratic form, of a Lie algebra a Chevalley–Eilenberg complex, of a contact algebra a cyclic A_{∞} -structure, of a mixed algebra an infinite L_{∞} -tower with no polynomial truncation.

The programme extends to non-principal $\mathcal{W}$ -algebras: the Bershadsky–Polyakov algebra BP_k is chirally Koszul at generic level ($k \neq -3$) with dual BP_{-k-6} and Koszul conductor $K_{\text{BP}} = 196$. Nilpotent transport from hook-type seeds covers all orbits in $\mathfrak{sl}_2$ through $\mathfrak{sl}_8$.

9.4. The PBW spectral sequence

The computational engine for bar cohomology within fixed genus is the PBW (Poincaré–Birkhoff–Witt) spectral sequence. Filter $\bar{B}^{(g)}(\mathcal{A})$ by *conformal weight*: assign each normally ordered monomial its total weight (sum of constituent field weights). The associated graded is the bar complex of a free-field algebra, reducing to symmetric-function combinatorics.

The E_1 page at genus g :

$$E_1^{p,q}(g) = E_1^{p,q}(g=0) \oplus \mathcal{E}_g^{p,q},$$

with the first summand the genus-0 bar cohomology determined by Koszulness of the associated graded and $\mathcal{E}_g^{p,q}$ the genus- g enrichment from the Hodge bundle, arising from the regular Eisenstein series coupled to the curvature. *Concentration*: $E_\infty^{p,q}(g) = 0$ for $q \neq 0$ at all $g \geq 1$, for all standard families. PBW degeneration: the spectral sequence collapses at E_2 and bar cohomology concentrates on the diagonal.

The PBW spectral sequence is distinct from the genus spectral sequence of §5.5. PBW uses conformal weight within fixed genus to compute bar cohomology groups; the genus spectral sequence uses the genus filtration on the deformation algebra to control the MC lift across genera. One computes *what*; the other determines *whether*.

9.5. Chiral Hochschild cohomology (Theorem H)

The chiral Hochschild complex $\text{ChirHoch}^\bullet(\mathcal{A})$ is the derived endomorphism algebra of the identity functor on $\mathcal{A}$ -modules, computed by the bar construction with bimodule coefficients. At genus 0,

$$\text{ChirHoch}^\bullet(\mathcal{A}) \simeq R\text{Hom}_{\mathcal{A}\text{-bimod}}(\mathcal{A}, \mathcal{A}) \simeq \text{Ext}_{\mathcal{A}^e}^\bullet(\mathcal{A}, \mathcal{A}),$$

where $\mathcal{A}^e = \mathcal{A} \otimes \mathcal{A}^{\text{op}}$ is the enveloping chiral algebra.

For $\mathcal{A} = \mathcal{W}^k(\mathfrak{g}, f_{\text{prin}})$ a principal $\mathcal{W}$ -algebra at generic level, the chiral Hochschild cohomology is concentrated in degrees $\{0, 1, 2\}$:

$$\text{ChirHoch}^n(\mathcal{W}^k(\mathfrak{g})) = 0 \text{ for } n > 2, \quad P(t) = 1 + t^2.$$

The deformation class in $\text{ChirHoch}^2 = \mathbb{C}$ is the level deformation $k \mapsto k + \epsilon$. The concentration bound is the de Rham amplitude on a curve ($\dim X = 1$); the continuous Lie algebra cohomology $H_{\text{cont}}^*(\text{Lie}(\mathcal{W}), \text{Lie}(\mathcal{W})_0; \mathbb{C}) = \mathbb{C}[\Theta_1, \dots, \Theta_r]$ (Gelfand–Fuchs, $\deg \Theta_i = m_i + 1$) is a different, unbounded invariant. For $\mathfrak{g} = \mathfrak{sl}_2$ (Virasoro): $P(t) = 1 + t^2$. For $\mathfrak{g} = \mathfrak{sl}_3$ ($\mathcal{W}_3$): $P(t) = 1 + t^2$. Total dimension ≤ 4 , never $\mathbb{C}[\Theta]$.

On the Koszul locus, $\text{ChirHoch}^\bullet(\mathcal{A})$ carries an E_2 -algebra structure that is formal. The Connes periodicity operator $S: \text{ChirHoch}^n \rightarrow \text{ChirHoch}^{n-2}$ intertwines Koszul duality: $S \circ \text{KD} = \text{KD} \circ S$. At genus 0, the chiral Hochschild complex identifies with the bulk chiral homology complex of Volume II. The bulk-boundary-line triangle is the genus-0 shadow of this identification: $\mathcal{A}_{\text{bulk}} \simeq Z_{\text{der}}(\mathcal{B}_\partial) \simeq \text{ChirHoch}^\bullet(\mathcal{A}^!)$.

9.6. The shadow algebra

The *shadow algebra* $\mathcal{A}^{\text{sh}}$ is the cohomology of the modular cyclic deformation complex as a graded-commutative ring,

$$\mathcal{A}^{\text{sh}} := H_\bullet(\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A})) = \bigoplus_{r \geq 2, g \geq 0} \mathcal{A}_{r,g}^{\text{sh}}.$$

The single MC element $\Theta_{\mathcal{A}}$ decomposes by degree and genus. Named shadows are projections:

$$\kappa(\mathcal{A}) = \mathcal{A}_{2,0}^{\text{sh}}, \quad \mathfrak{C}(\mathcal{A}) = \mathcal{A}_{3,0}^{\text{sh}}, \quad \Omega(\mathcal{A}) = \mathcal{A}_{4,0}^{\text{sh}}, \quad \text{Sh}_r(\mathcal{A}) = \mathcal{A}_{r,0}^{\text{sh}} \text{ for } r \geq 5.$$

The shadow algebra organises the entire hierarchy; it is a homotopy invariant, computed by the Weil homomorphism of the Chern–Weil dictionary (§5.4).

The all-degree master equation in the shadow algebra: $\nabla_H(\text{Sh}_r) + o^{(r)} = 0$ for all $r \geq 3$. The obstruction class $o^{(r)}$ lies in $\mathcal{A}_{r,0}^{\text{sh}}$ and measures the failure of $\Theta_{\mathcal{A}}^{\leq r-1}$ to extend to degree r .

9.7. BV/BRST identification at genus 0

At genus 0, the bar differential of a chiral algebra on $\mathbb{C}$ is the BRST differential of the associated holomorphic twist, $(d_{\bar{B}})_{g=0} = Q_{\text{BRST}}$. The bar complex $\bar{B}_{\mathbb{C}}^{(0)}(\mathcal{A})$ is the BRST complex of the 2d chiral algebra, and bar-cobar inversion $\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ (Theorem B) says the BRST cohomology processed through the cobar functor reconstructs the original algebra.

At genus 1, the identification acquires curvature: $d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_1$ is the one-loop anomaly, and the curved bar complex is the curved BRST complex of the theory on an elliptic curve. The complementarity $Q_1(\mathcal{A}) \oplus Q_1(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_{1,1}, \mathcal{Z}_{\mathcal{A}})$ is the genus-1 anomaly cancellation condition.

At genus $g \geq 2$, the identification with Costello–Gwilliam requires renormalisation: higher-loop integrals over $\overline{\mathcal{M}}_{g,n}$ need regularisation beyond the algebraic bar-cobar framework, and the correct receptacle is the coderived category.

For the Heisenberg algebra, the identification is proved at all genera by four independent paths: zeta-regularisation, Selberg zeta function, Grothendieck–Riemann–Roch, direct bar computation. The shadow obstruction tower equals the L_∞ -formality obstruction tower at all degrees.

Genus-two shell activation as depth diagnostic. The genus-two shell profile independently confirms the depth classification:

	$\Theta_{\text{loop}^2}^{(2)}$	$\Theta_{\text{sepoloop}}^{(2)}$	$\Theta_{\text{pf}}^{(2)}$
G (Heisenberg, lattice)	✓	—	—
L (affine $\widehat{\mathfrak{g}}_k$)	✓	✓	—
C ($\beta\gamma$)	✓	—	—
M (Virasoro, $\mathcal{W}_N$)	✓	✓	✓

G and C activate only the looploop shell; L adds sepoloop; only M activates all three including planted-forest. The planted-forest shell is the diagnostic: its presence forces infinite shadow depth.

Structural refinements. The three bar complexes on a chiral algebra (ordered tensor T^c , symmetric Sym^c , Chevalley–Eilenberg Lie^c) sit in a chain $\text{Lie}^c \hookrightarrow \text{Sym}^c \hookrightarrow T^c$ with distinct coalgebra structures at each level (Theorem ??). The averaging map $\text{av}: T^c \rightarrow \text{Sym}^c$ from ordered to symmetric is surjective but not split: the kernel at degree 3 contains the Drinfeld associator $\Phi_{\text{KZ}}(\mathcal{A})$, and the fibre of splittings is a torsor for GRT_1 . The E_1 ordered bar complex is therefore the natural primitive object; the modular/symmetric story is its Σ_n -coinvariant image (Theorem ??). The genus-3 cross-channel correction for $\mathcal{W}_3$ has been computed by summing over all 42 stable graphs of $\overline{\mathcal{M}}_{3,0}$ (Proposition ??): the cross-channel tower accelerates with genus and dominates the scalar part $\kappa \cdot \lambda_g^{\text{FP}}$ by genus 4.

On the open/closed side, the Koszul dual cooperad $\text{SC}^{\text{ch}, \text{top}, !}$ decomposes into three sectors: closed ($\dim = (n-1)!$, Lie cooperad), open ($\dim = m!$, Ass cooperad), and mixed ($\dim = (k-1)! \binom{k+m}{m}$). The mixed sector encodes the bulk-to-boundary module structure by which the chiral derived centre acts on the boundary algebra.

The BV/BRST = bar identification at higher genus (Conjecture ??): the chain-level obstruction (harmonic propagator correction) is resolved for classes G, L, and C, for G and L by the Heisenberg sewing theorem and its extension to affine algebras, for C by contact decoupling. For class M (Virasoro, $\mathcal{W}_N$) the identification holds on bar cohomology but fails at the chain level: the harmonic propagator produces a nonvanishing quartic obstruction, so a coderived reformulation is needed.